Visualization with ggplot2

Lecture 10

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ggplot2

www.rstudio.com

The Grammar of Graphics

- Visualization concept created by Leland Wilkinson (The Grammar of Graphics, 1999)
- attempt to taxonomize the basic elements of statistical graphics
- Adapted for R by Hadley Wickham (2009)
  - consistent and compact syntax to describe statistical graphics
  - highly modular as it breaks up graphs into semantic components
  - ggplot2 is not meant as a guide to which graph to use and how to best convey your data (more on that later), but it does have some strong opinions.

Terminology

A statistical graphic is a…

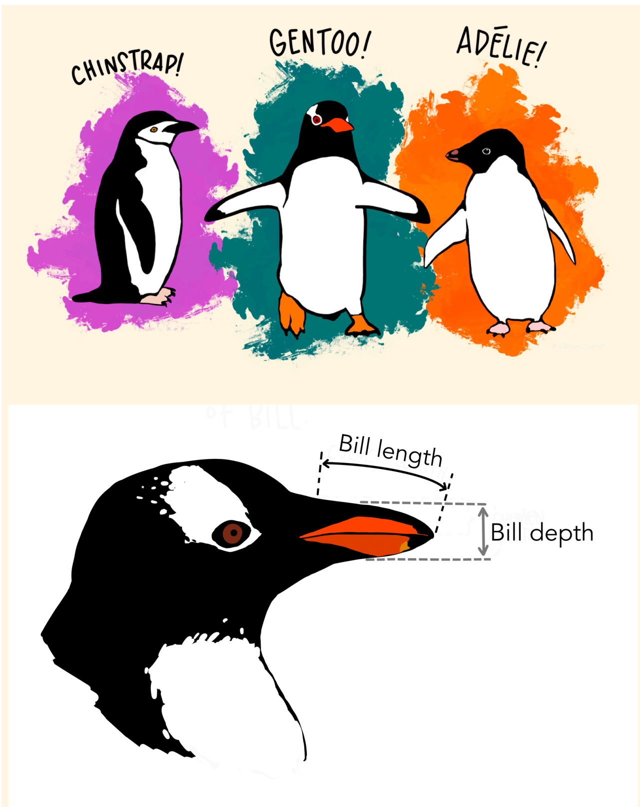
- mapping of **data**
- which may be **statistically transformed** (summarized, log-transformed, etc.)
- to **aesthetic attributes** (color, size, xy-position, etc.)
- using **geometric objects** (points, lines, bars, etc.)
- and mapped onto a specific **facet** and **coordinate system**

Anatomy of a ggplot call

```
ggplot(
  data = [dataframe],
  mapping = aes(
    x = [var x], y = [var y],
    color = [var color],
    shape = [var shape],
    ...
  )
) +
  geom_[some geom](
    mapping = aes(
      color = [var geom color],
      ...
    )
  ) +
  ... # other geometries
  scale_[some axis]_[some scale]() +
  facet_[some facet]([formula]) +
  ... # other options
```

Data - Palmer Penguins

Measurements for penguin species, island in Palmer Archipelago, size (flipper length, body mass, bill dimensions), and sex.




```r
library(palmerpenguins)
penguins
```

```
# A tibble: 344 × 8
   species island    bill_length_mm bill_depth_mm
   <fct>   <fct>              <dbl>         <dbl>
 1 Adelie  Torgersen           39.1          18.7
 2 Adelie  Torgersen           39.5          17.4
 3 Adelie  Torgersen           40.3          18  
 4 Adelie  Torgersen           NA            NA  
 5 Adelie  Torgersen           36.7          19.3
 6 Adelie  Torgersen           39.3          20.6
 7 Adelie  Torgersen           38.9          17.8
 8 Adelie  Torgersen           39.2          19.6
 9 Adelie  Torgersen           34.1          18.1
10 Adelie  Torgersen           42            20.2
# ℹ 334 more rows
# ℹ 4 more variables: flipper_length_mm <int>,
#   body_mass_g <int>, sex <fct>, year <int>
```

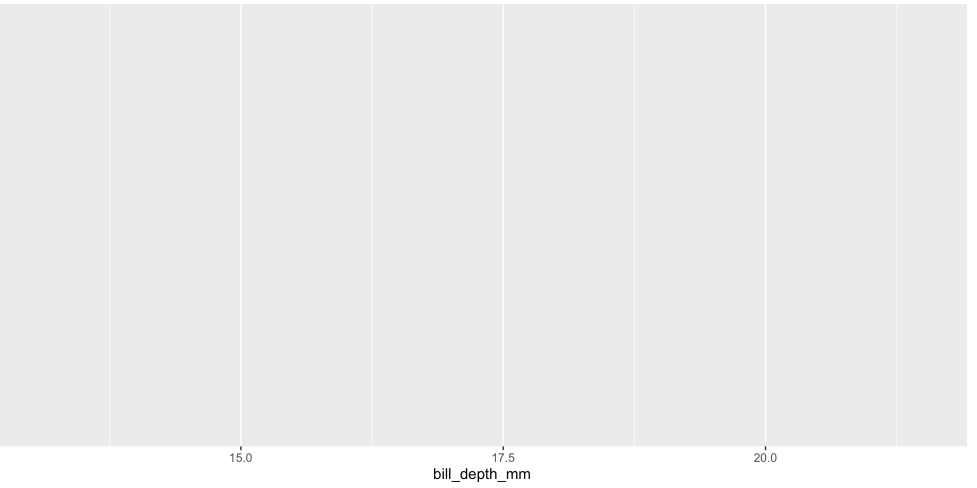
Text <-> Plot

> *Start with the `penguins` data frame*

```
ggplot(data = penguins)
```

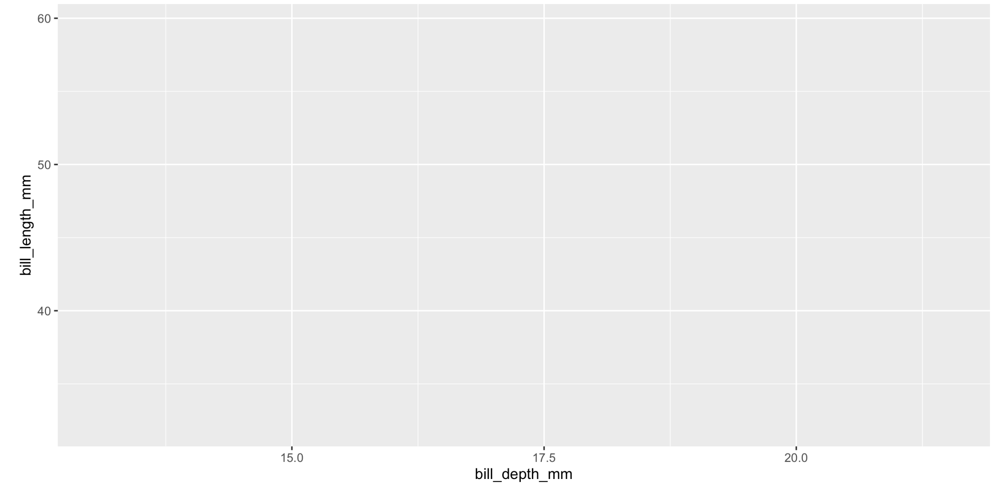
> Start with the `penguins` data frame, *map bill depth to the x-axis*

```r
ggplot(
  data = penguins,
  mapping = aes(
    x = bill_depth_mm
  )
)
```



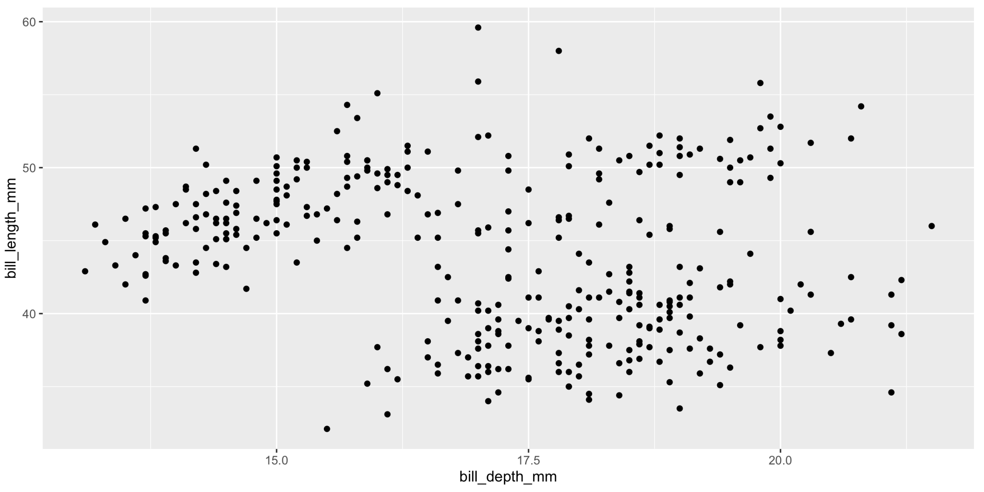
> Start with the `penguins` data frame, map bill depth to the x-axis *and map bill length to the y-axis.*

```r
ggplot(
  data = penguins,
  mapping = aes(
    x = bill_depth_mm,
    y = bill_length_mm
  )
)
```



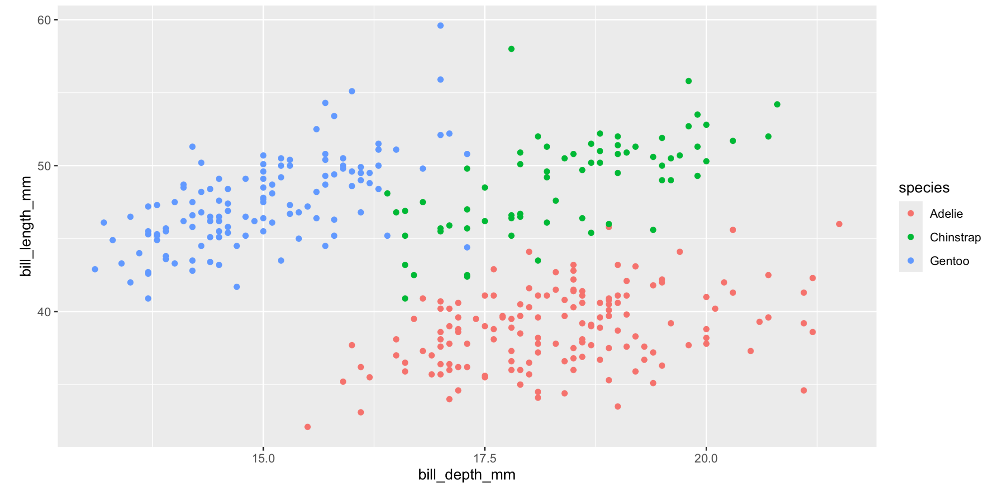
> Start with the `penguins` data frame, map bill depth to the x-axis and map bill length to the y-axis. *Represent each observation with a point*

```
ggplot(
  data = penguins,
  mapping = aes(
    x = bill_depth_mm,
    y = bill_length_mm
  )
) +
  geom_point()
```



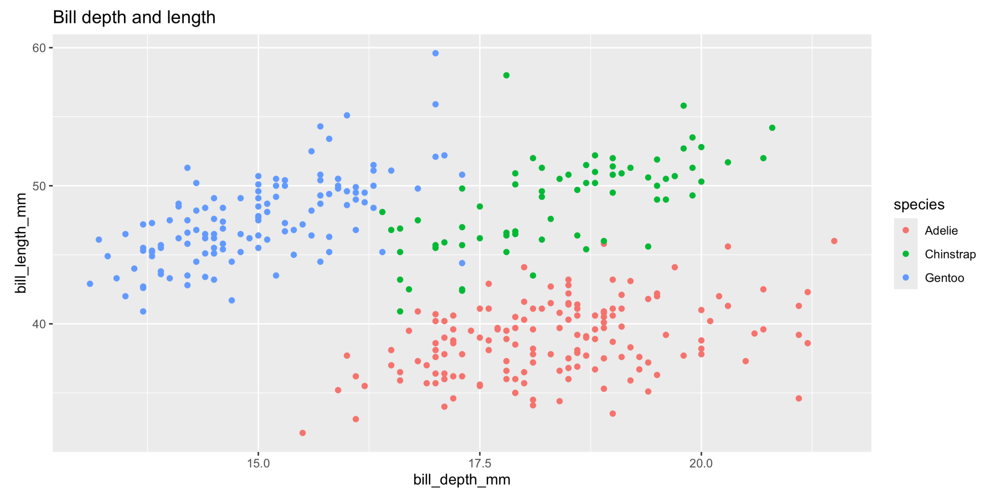
> Start with the `penguins` data frame, map bill depth to the x-axis and map bill length to the y-axis. Represent each observation with a point *and map species to the color of each point.*

```r
ggplot(
  data = penguins,
  mapping = aes(
    x = bill_depth_mm,
    y = bill_length_mm
  )
) +
  geom_point(
    mapping = aes(color = species)
  )
```



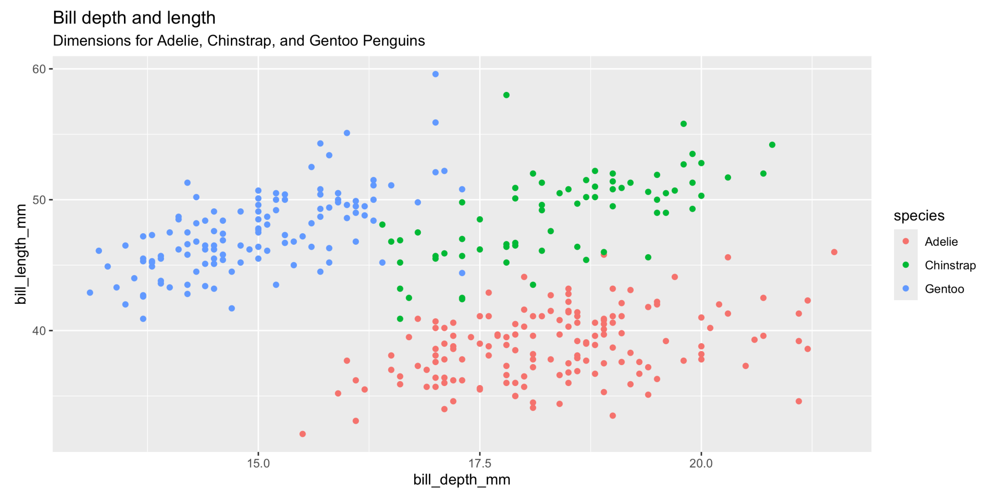
> Start with the `penguins` data frame, map bill depth to the x-axis and map bill length to the y-axis. Represent each observation with a point and map species to the color of each point. *Title the plot "Bill depth and length"*

```r
ggplot(
  data = penguins,
  mapping = aes(
    x = bill_depth_mm,
    y = bill_length_mm
  )
) +
  geom_point(
    mapping = aes(color = species)
  ) +
  labs(title = "Bill depth and length")
```



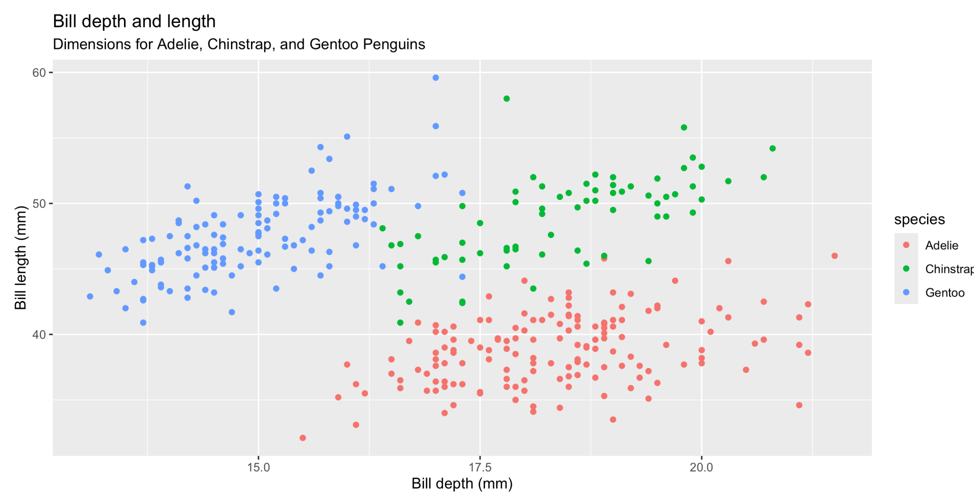
> Start with the `penguins` data frame, map bill depth to the x-axis and map bill length to the y-axis. Represent each observation with a point and map species to the color of each point. Title the plot “Bill depth and length”, *add the subtitle “Dimensions for Adelie, Chinstrap, and Gentoo Penguins”*

```r
ggplot(
  data = penguins,
  mapping = aes(
    x = bill_depth_mm,
    y = bill_length_mm
  )
) +
  geom_point(
    mapping = aes(color = species)
  ) +
  labs(
    title = "Bill depth and length",
    subtitle = paste("Dimensions for Adelie
                     "Chinstrap, and Gentoo'
                     "Penguins")
  )
```



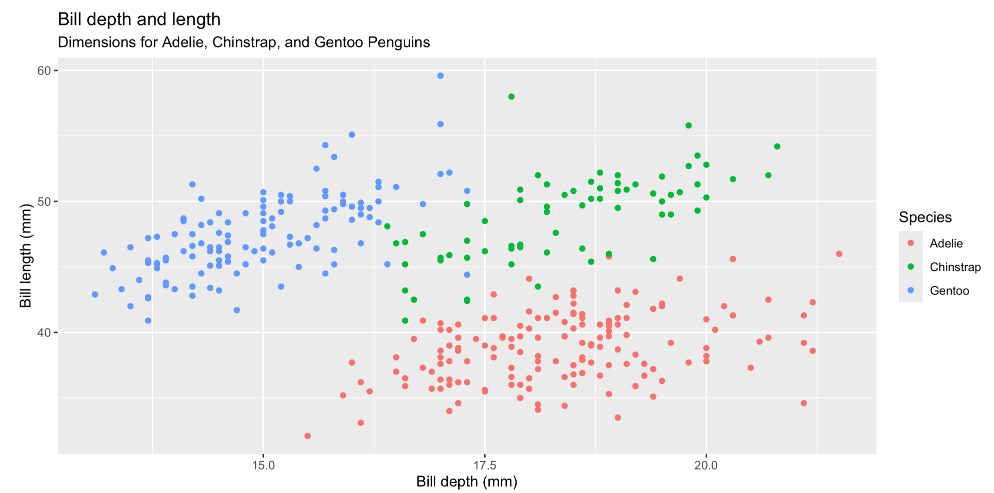
> Start with the `penguins` data frame, map bill depth to the x-axis and map bill length to the y-axis. Represent each observation with a point and map species to the color of each point. Title the plot “Bill depth and length”, add the subtitle “Dimensions for Adelie, Chinstrap, and Gentoo Penguins”, *label the x and y axes as “Bill depth (mm)” and “Bill length (mm)”, respectively*

```r
ggplot(
  data = penguins,
  mapping = aes(
    x = bill_depth_mm,
    y = bill_length_mm
  )
) +
  geom_point(
    mapping = aes(color = species)
  ) +
  labs(
    title = "Bill depth and length",
    subtitle = paste("Dimensions for Adelie,
                     "Chinstrap, and Gentoo"
                     "Penguins"),
    x = "Bill depth (mm)",
    y = "Bill length (mm)"
  )
```



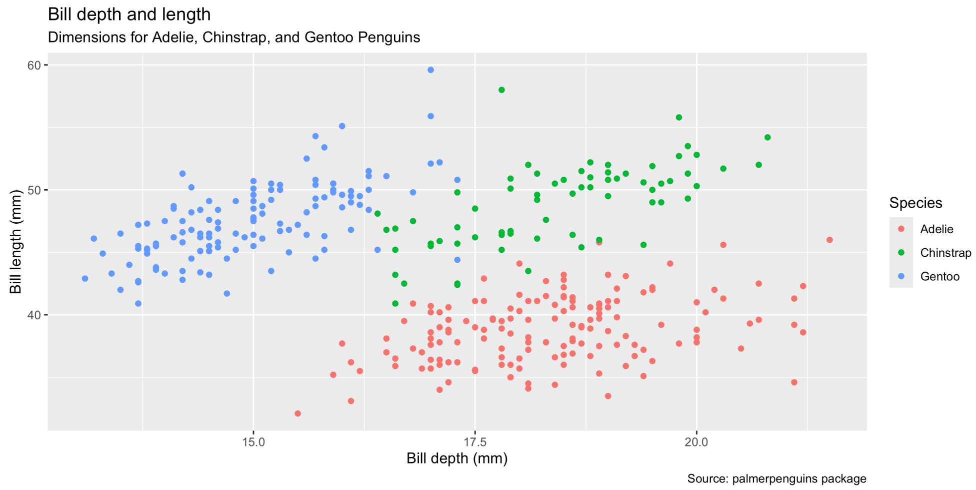
> Start with the `penguins` data frame, map bill depth to the x-axis and map bill length to the y-axis. Represent each observation with a point and map species to the color of each point. Title the plot “Bill depth and length”, add the subtitle “Dimensions for Adelie, Chinstrap, and Gentoo Penguins”, label the x and y axes as “Bill depth (mm)” and “Bill length (mm)”, respectively, *label the legend “Species”*

```r
ggplot(
  data = penguins,
  mapping = aes(
    x = bill_depth_mm,
    y = bill_length_mm
  )
) +
  geom_point(
    mapping = aes(color = species)
  ) +
  labs(
    title = "Bill depth and length",
    subtitle = paste("Dimensions for Adelie
                     "Chinstrap, and Gentoo
                     "Penguins"),
    x = "Bill depth (mm)",
    y = "Bill length (mm)",
    color = "Species"
  )
```



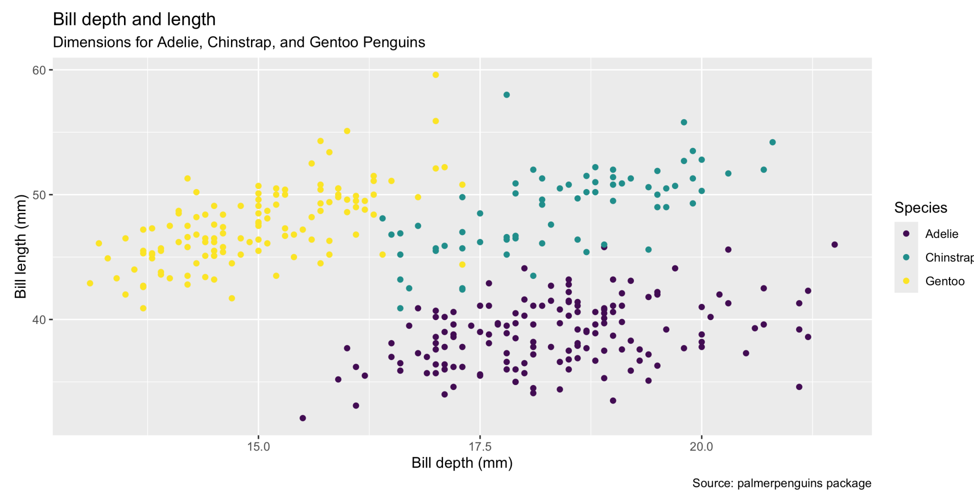
> Start with the `penguins` data frame, map bill depth to the x-axis and map bill length to the y-axis. Represent each observation with a point and map species to the color of each point. Title the plot "Bill depth and length", add the subtitle "Dimensions for Adelie, Chinstrap, and Gentoo Penguins", label the x and y axes as "Bill depth (mm)" and "Bill length (mm)", respectively, label the legend "Species", *and add a caption for the data source.*

```r
ggplot(
  data = penguins,
  mapping = aes(
    x = bill_depth_mm,
    y = bill_length_mm
  )
) +
  geom_point(
    mapping = aes(color = species)
  ) +
  labs(
    title = "Bill depth and length",
    subtitle = paste("Dimensions for Adelie
                     "Chinstrap, and Gentoo
                     "Penguins"),
    x = "Bill depth (mm)",
    y = "Bill length (mm)",
    color = "Species",
    caption = "Source: palmerpenguins packa
  )
```



> Start with the `penguins` data frame, map bill depth to the x-axis and map bill length to the y-axis. Represent each observation with a point and map species to the color of each point. Title the plot “Bill depth and length”, add the subtitle “Dimensions for Adelie, Chinstrap, and Gentoo Penguins”, label the x and y axes as “Bill depth (mm)” and “Bill length (mm)”, respectively, label the legend “Species”, and add a caption for the data source. *Finally, use the viridis color palette for all points.*

```r
ggplot(
    data = penguins,
    mapping = aes(
      x = bill_depth_mm,
      y = bill_length_mm
    )
  ) +
    geom_point(
      mapping = aes(color = species)
    ) +
    labs(
      title = "Bill depth and length",
      subtitle = paste("Dimensions for Adeli
                       "Chinstrap, and Gento
                       "Penguins"),
      x = "Bill depth (mm)",
      y = "Bill length (mm)",
      color = "Species",
      caption = "Source: palmerpenguins pack
    ) +
    scale_color_viridis_d()
```



Aesthetics

Aesthetics options

Commonly used characteristics of plotting geometries that can be **mapped to a specific variable** in the data, examples include:

- position (`x`, `y`)
- `color`
- `shape`
- `size`
- `alpha` (transparency)

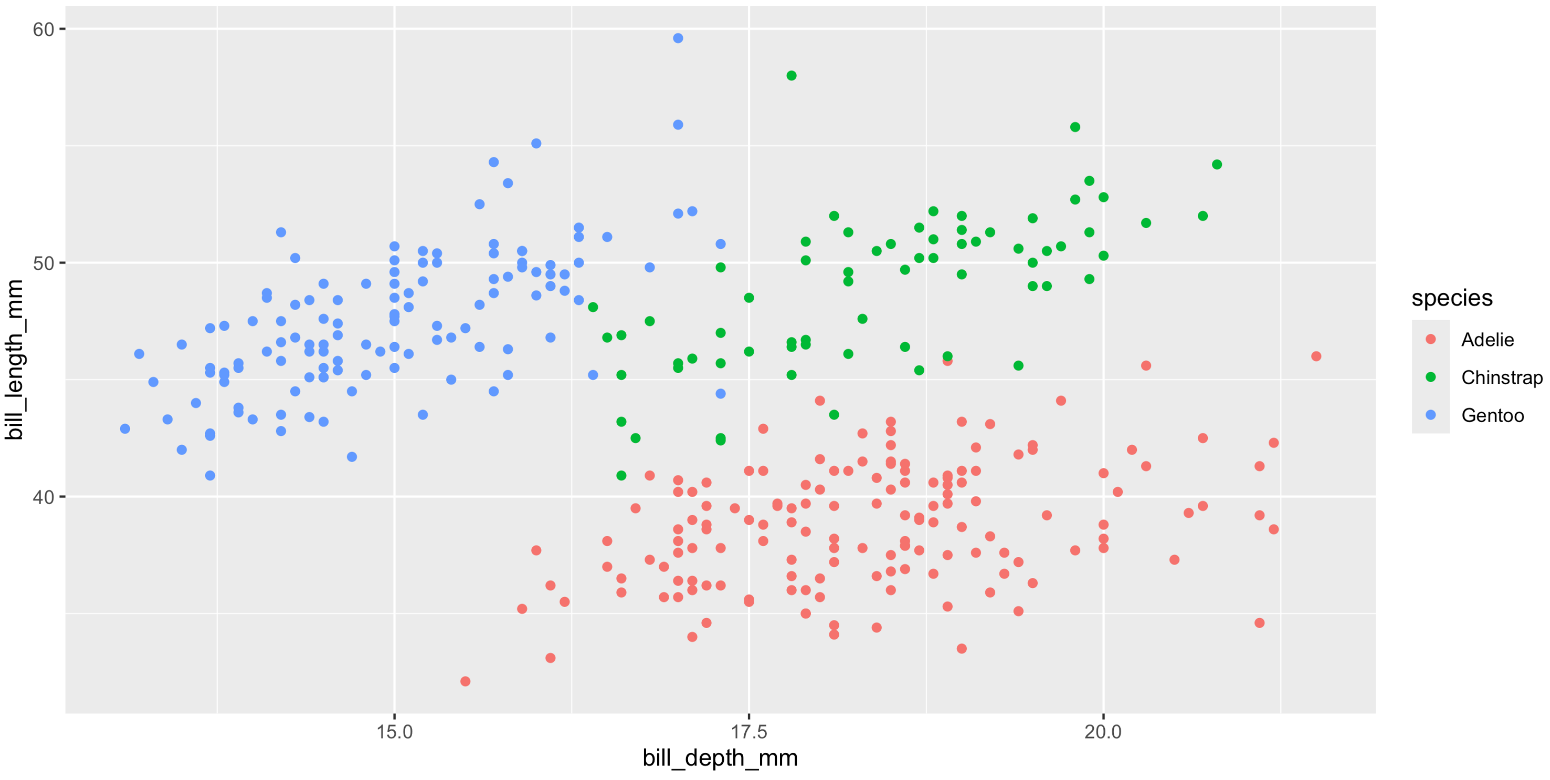
Different geometries have different aesthetics available - see the ggplot2 geoms help files for listings.

- Aesthetics given in `ggplot()` apply to all `geom`s.
- Aesthetics for a specific `geom_*()` can be overridden via `mapping` or as an argument.

color

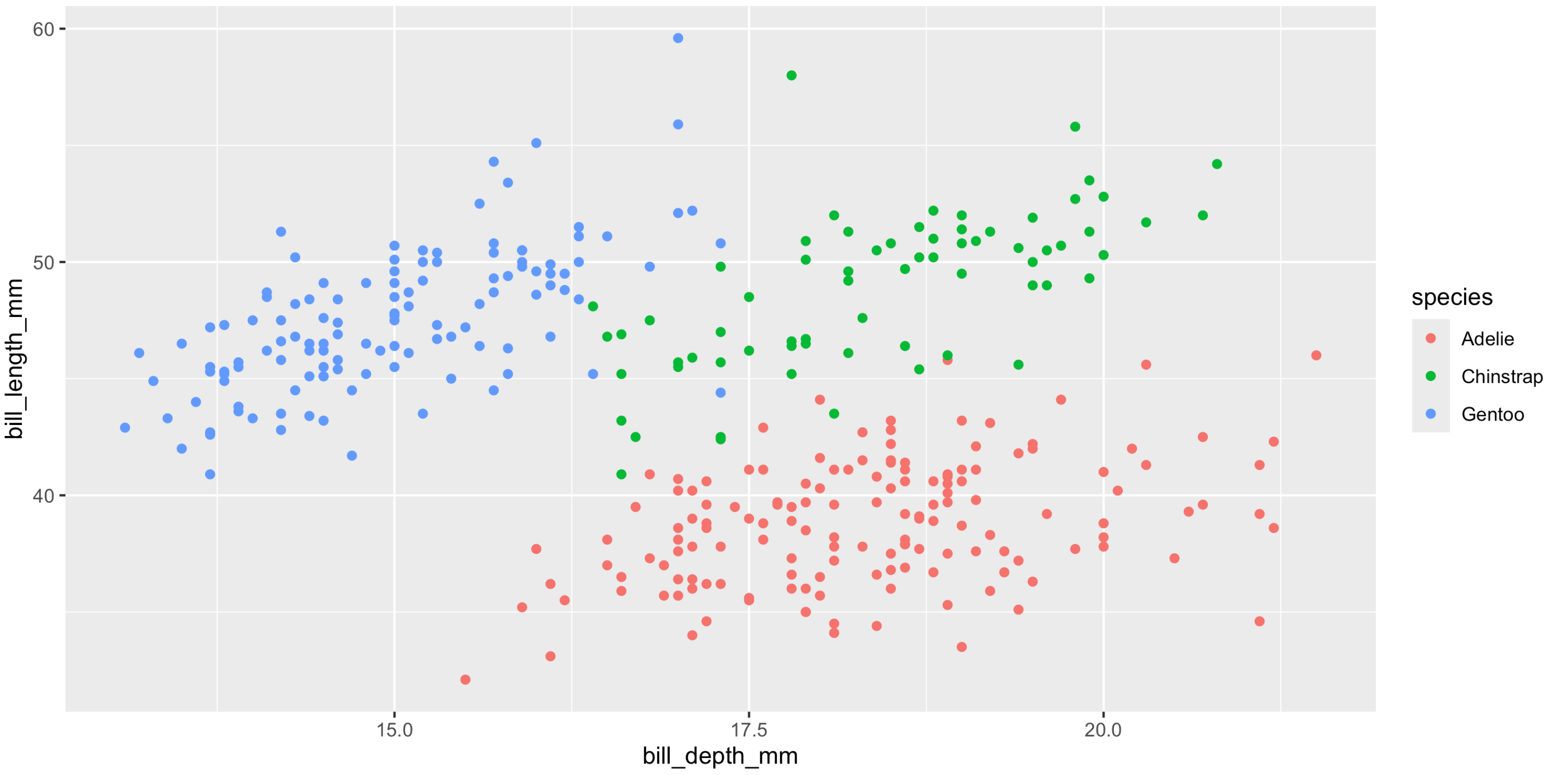
```
ggplot(
  penguins, aes(x = bill_depth_mm, y = bill_length_mm)
) +
  geom_point(
    aes(color = species)
  )
```

```
Warning: Removed 2 rows containing missing values or values outside the
scale range (`geom_point()`).
```



Stop the warning

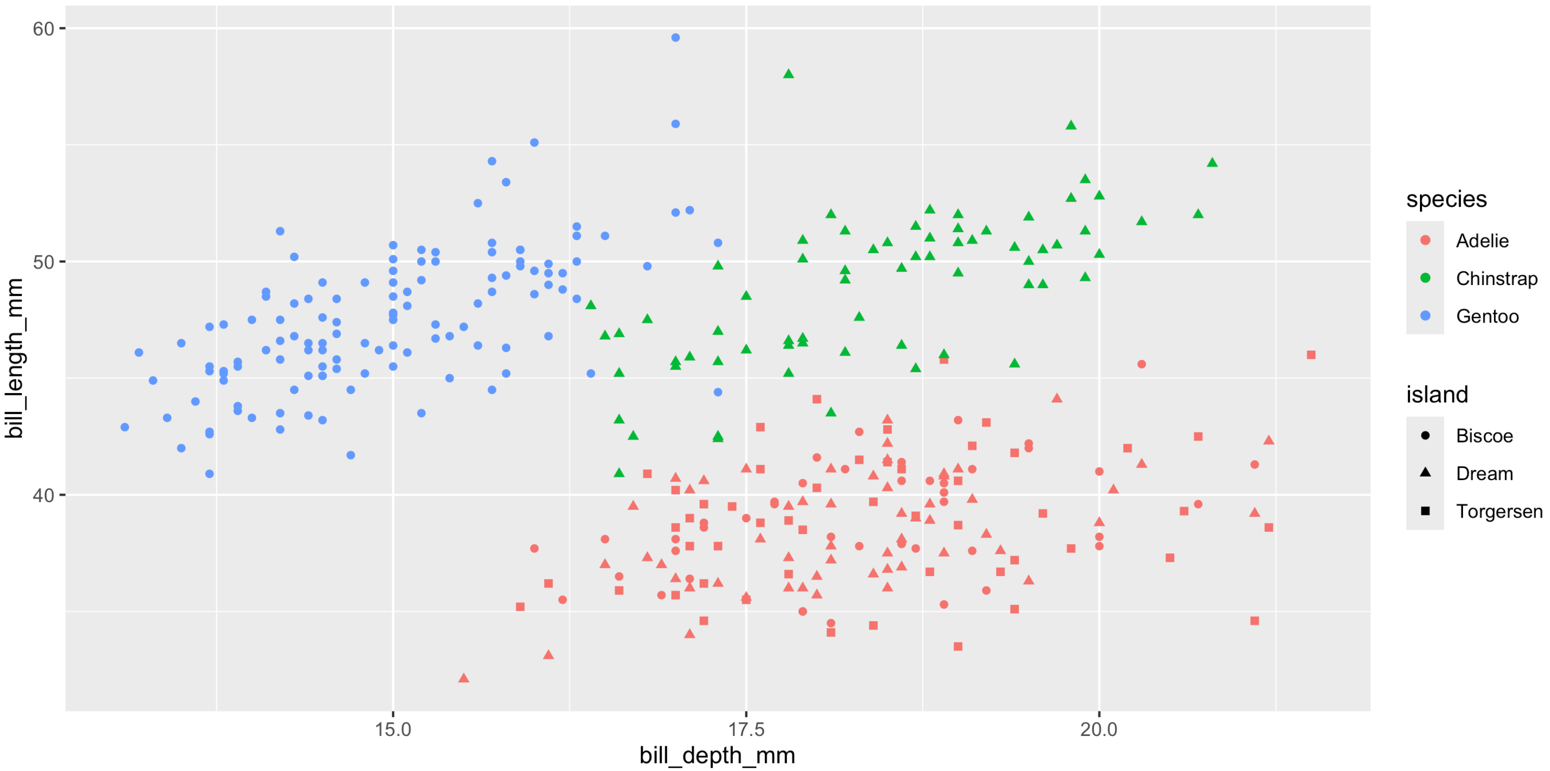
```
ggplot(
  penguins, aes(x = bill_depth_mm, y = bill_length_mm)
) +
  geom_point(
    aes(color = species), na.rm=TRUE
  )
```



Shape

Mapped to a different variable than `color`

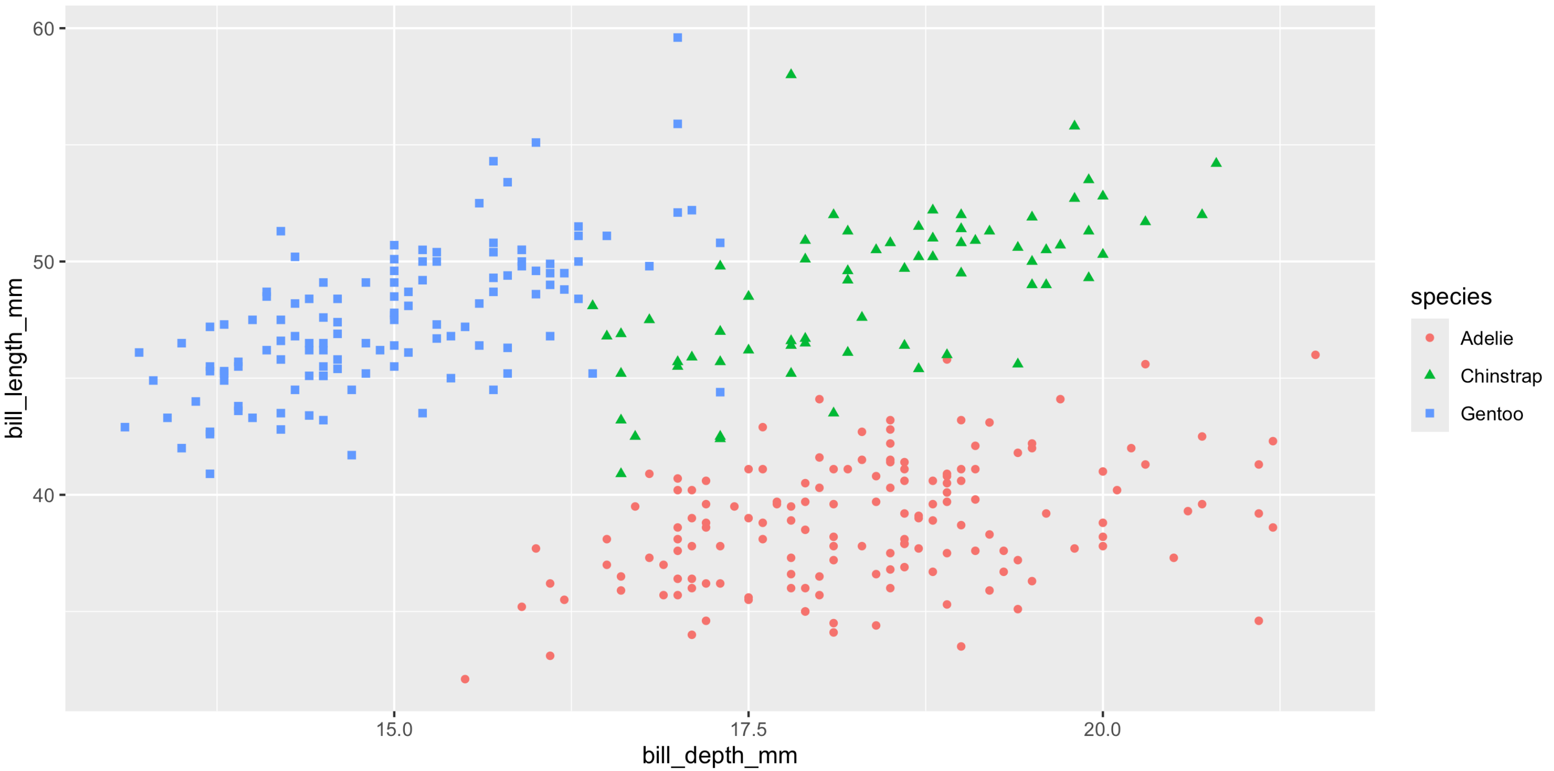
```
ggplot(
  penguins, aes(x = bill_depth_mm, y = bill_length_mm)
) +
  geom_point(
    aes(color = species, shape = island), na.rm = TRUE
  )
```



Shape

Mapped to same variable as `color`

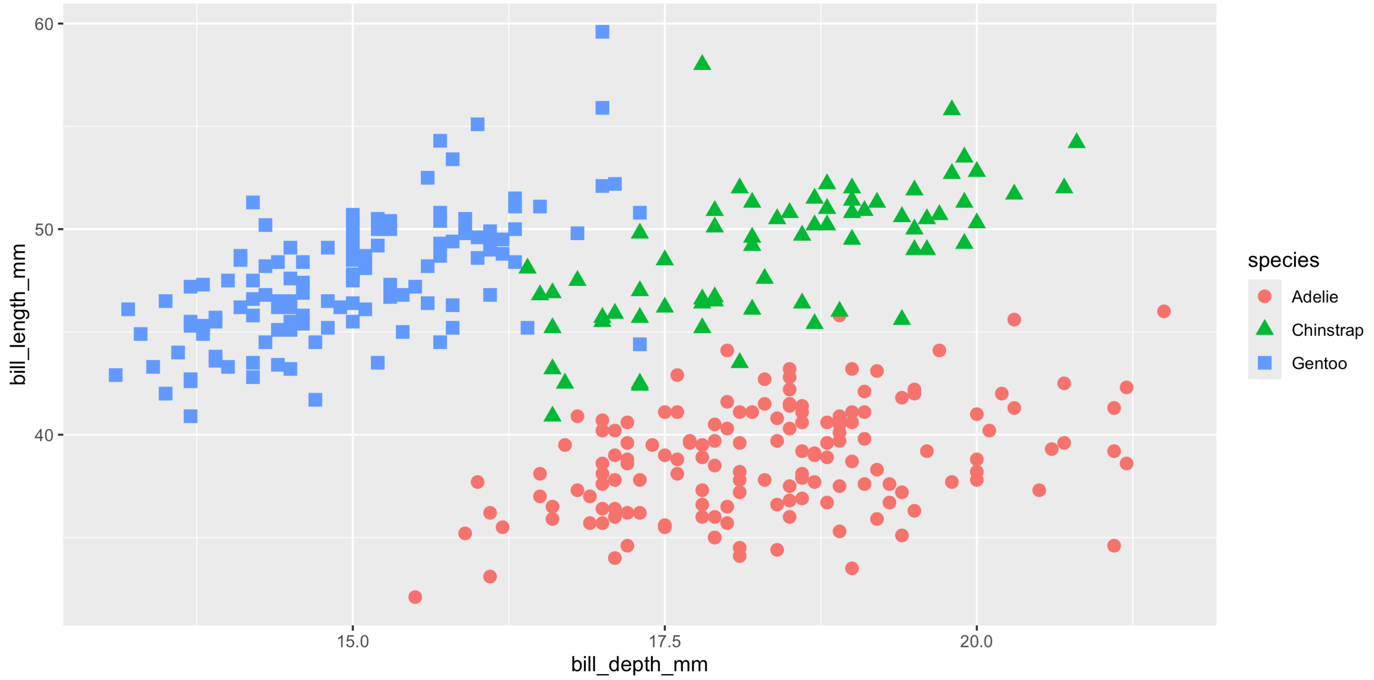
```
ggplot(
  penguins, aes(x = bill_depth_mm, y = bill_length_mm)
) +
  geom_point(
    aes(color = species, shape = species), na.rm = TRUE
  )
```



Size

Using a fixed value - note that this value is outside of the aes call

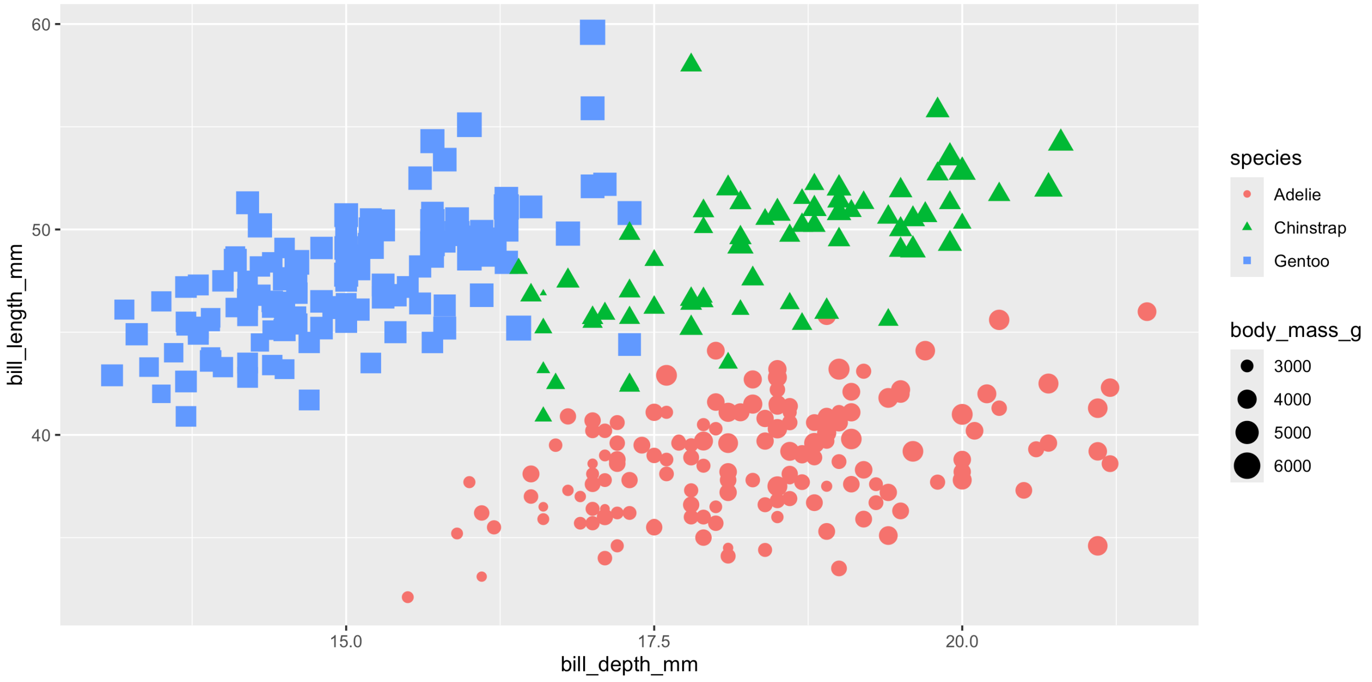
```
ggplot(
  penguins, aes(x = bill_depth_mm, y = bill_length_mm)
) +
  geom_point(
    aes(color = species, shape = species), na.rm = TRUE,
    size = 3
  )
```



Size

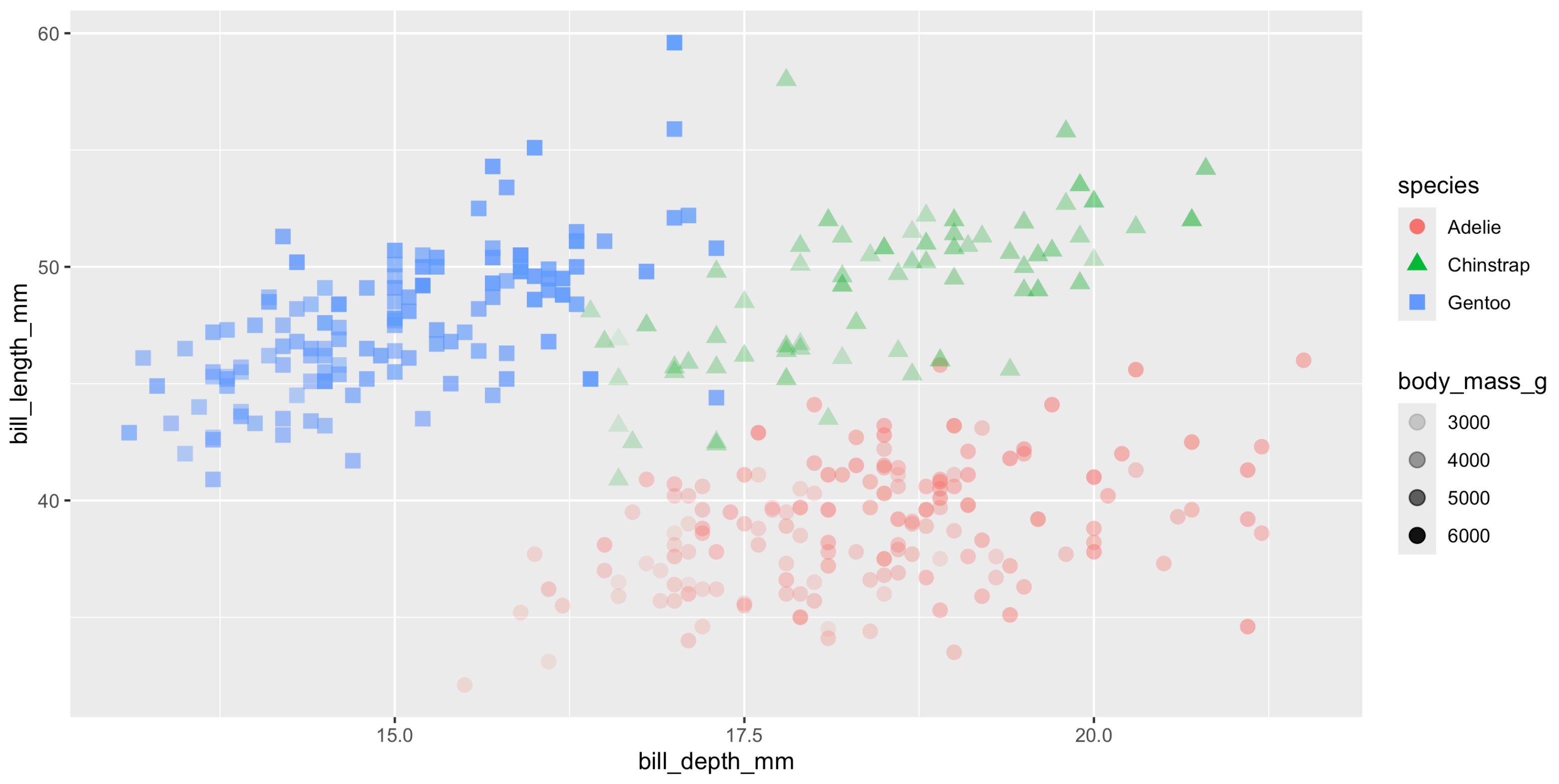
Mapped to a variable

```r
ggplot(
  penguins, aes(x = bill_depth_mm, y = bill_length_mm)
) +
  geom_point(
    aes(color = species, shape = species, size = body_mass_g), na.rm = TRUE
  )
```



Alpha

```r
ggplot(
  penguins,
  aes(x = bill_depth_mm, y = bill_length_mm)
) +
  geom_point(
    aes(color = species, shape = species, alpha = body_mass_g), na.rm = TRUE,
    size = 3
  )
```



Mapping vs settings

- **Mapping** - Determine an aesthetic (the size, alpha, etc.) of a geom based on the values of a variable in the data
  - wrapped by `aes()` and pass as `mapping` argument to `ggplot()` or `geom_*()`.

- **Setting** - Determine an aesthetic (the size, alpha, etc.) of a geom using a constant value not directly from the data.
  - passed directly into `geom_*()` as an argument.

From the previous slide `color`, `shape`, and `alpha` are all **aesthetics** while `size` was a **setting**.

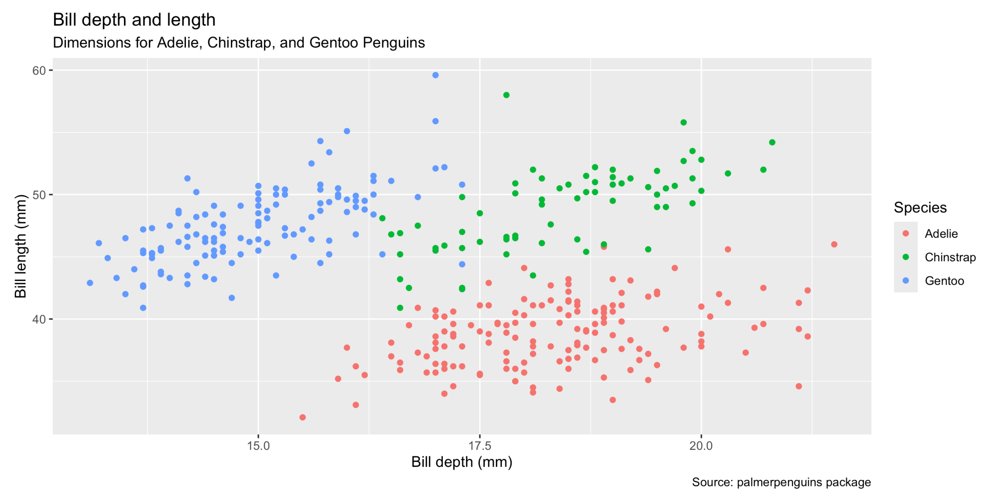
Labels

`labs()`

In our previous example we saw the use of `labs()` to human readable labels to various plot components.

We it comes to out aesthetics (data) we can annotate the data so that the label is generated automatically, by attaching a `label` argument to the appropriate column in our data frame.

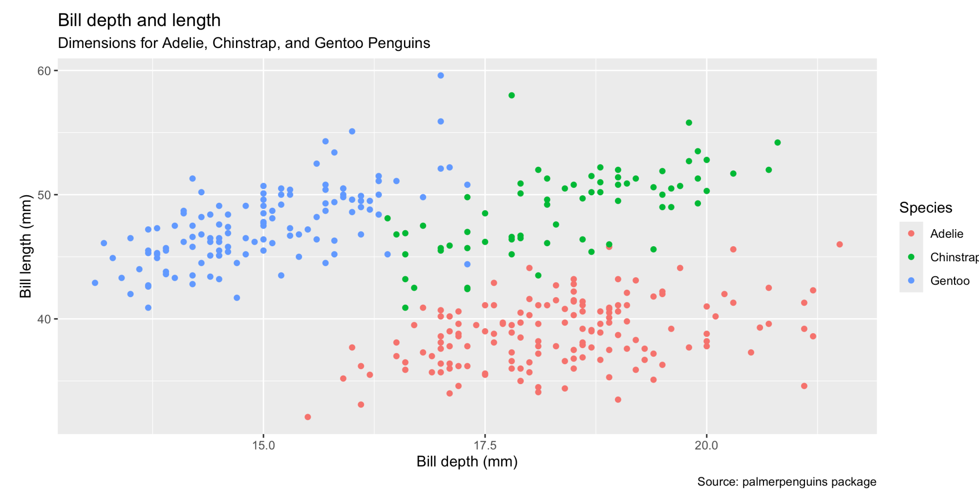
```r
ggplot(
  data = penguins,
  mapping = aes(
    x = bill_depth_mm,
    y = bill_length_mm
  )
) +
  geom_point(
    mapping = aes(color = species), na.rm = TRUE
  ) +
  labs(
    title = "Bill depth and length",
    subtitle = paste(
      "Dimensions for Adelie,",
      "Chinstrap, and Gentoo",
      "Penguins"),
    x = "Bill depth (mm)",
    y = "Bill length (mm)",
    color = "Species",
    caption = "Source: palmerpenguins package"
  )
```



Labels

```r
p_labeled = penguins
attr(p_labeled$species, "label") = "Species"
attr(p_labeled$bill_depth_mm, "label") = "Bill depth (mm)"
attr(p_labeled$bill_length_mm, "label") = "Bill length (mm

ggplot(
  data = p_labeled,
  mapping = aes(
    x = bill_depth_mm,
    y = bill_length_mm
  )
) +
  geom_point(
    mapping = aes(color = species), na.rm = TRUE
  ) +
  labs(
    title = "Bill depth and length",
    subtitle = paste(
      "Dimensions for Adelie,",
      "Chinstrap, and Gentoo",
      "Penguins"),
    caption = "Source: palmerpenguins package"
  )
```



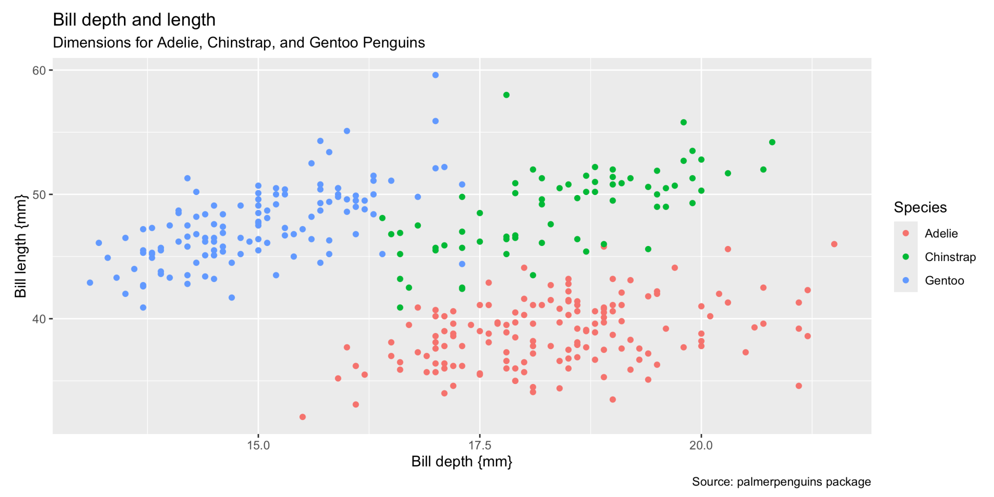
The benefit of this approach is that the labels are now part of the data frame and will be used automatically in future plots. Define once, use repeatedly.

Dictionary

Alternatively, we can provide a dictionary / lookup table to `labs()` dictionary, which will then be used.

```r
lookup = c(
  species = "Species",
  bill_depth_mm = "Bill depth {mm}",
  bill_length_mm = "Bill length {mm}"
)

ggplot(
  data = penguins,
  mapping = aes(
    x = bill_depth_mm,
    y = bill_length_mm
  )
) +
  geom_point(
    mapping = aes(color = species), na.rm = TRUE
  ) +
  labs(
    title = "Bill depth and length",
    subtitle = paste(
      "Dimensions for Adelie,",
      "Chinstrap, and Gentoo",
      "Penguins"),
    caption = "Source: palmerpenguins package",
    dictionary = lookup
  )
```



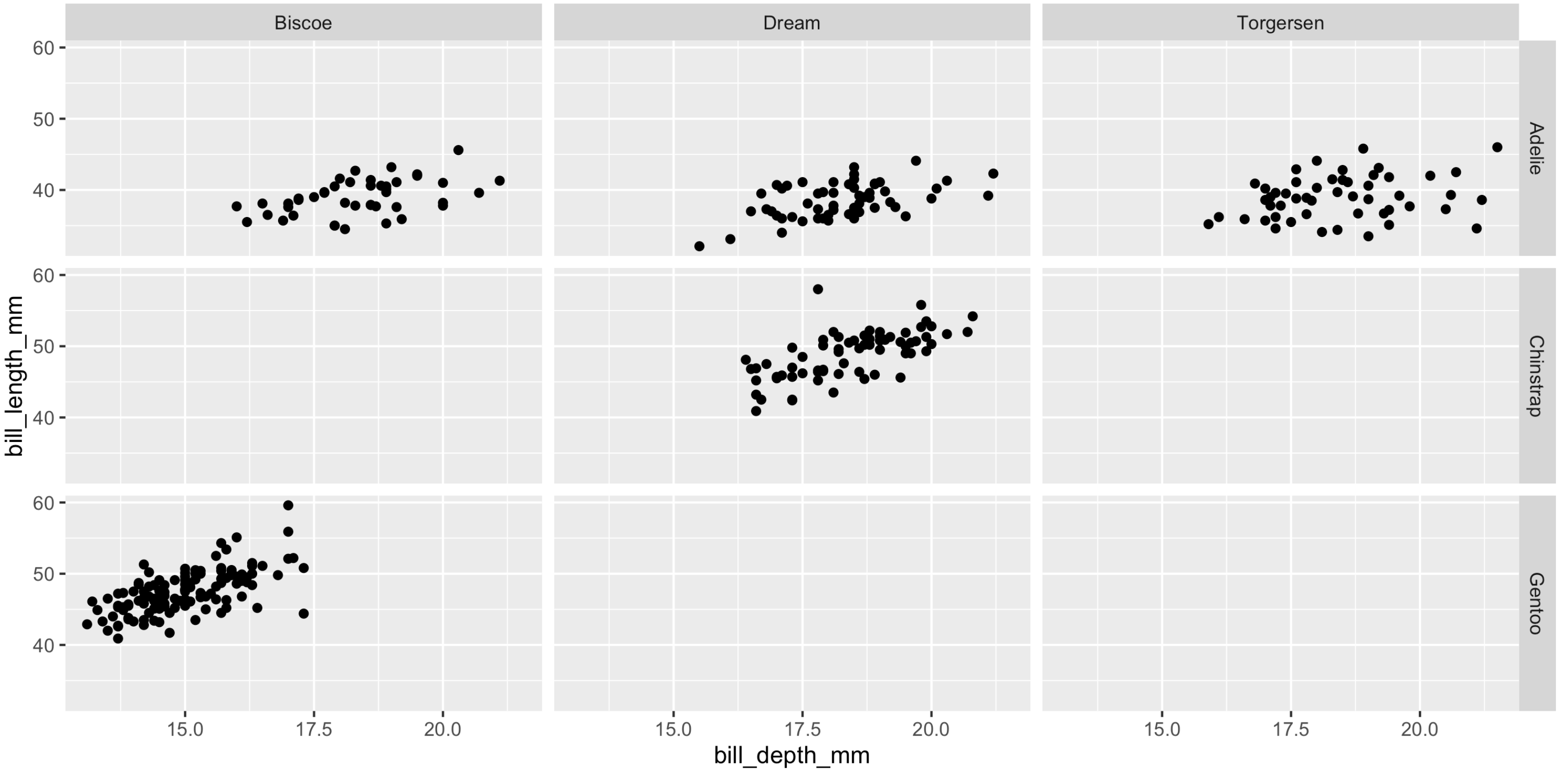
Faceting

Faceting

- Smaller plots that display different subsets of the data
- Useful for exploring conditional relationships and large data
- Sometimes referred to as “small multiples”

facet_grid

```
ggplot(
  penguins, aes(x = bill_depth_mm, y = bill_length_mm)
) +
  geom_point(na.rm = TRUE) +
  facet_grid(
    species ~ island
  )
```

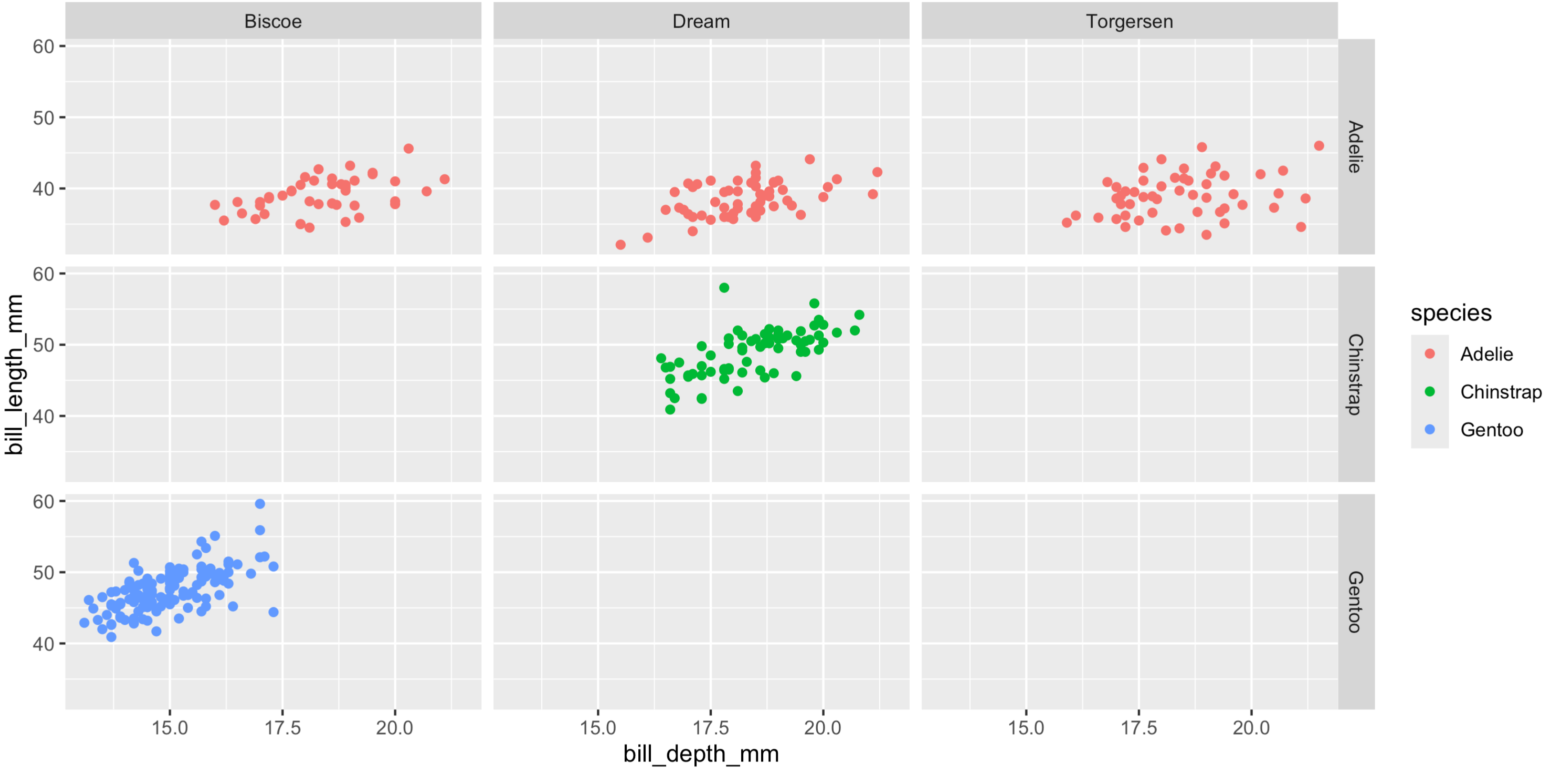


Compare with ...

```
ggplot(
  penguins, aes(x = bill_depth_mm, y = bill_length_mm)
) +
  geom_point(
    aes(color = species, shape = island), na.rm = TRUE, size = 3
  )
```

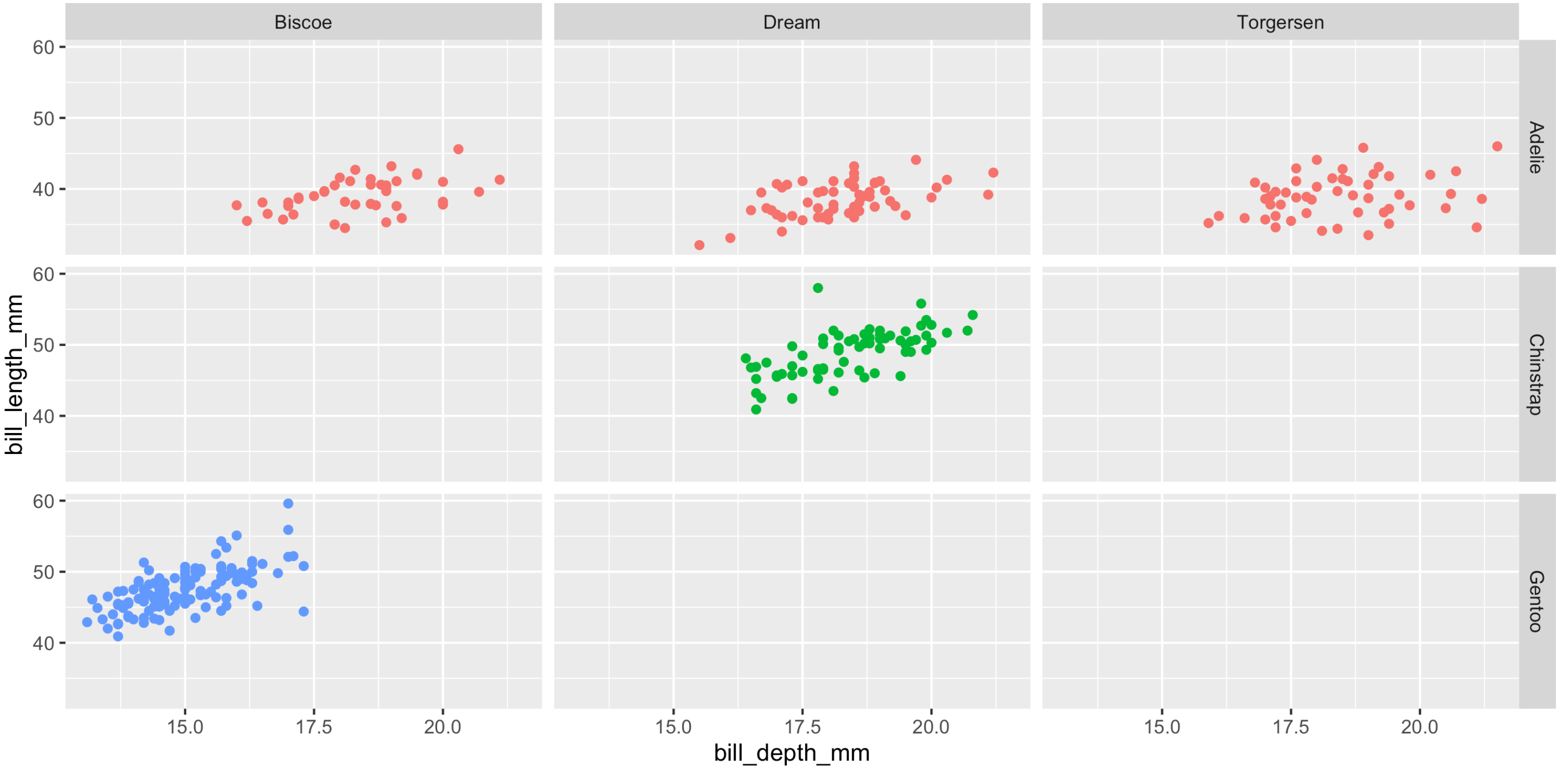
Faceting and color

```r
ggplot(
  penguins, aes(x = bill_depth_mm, y = bill_length_mm, color = species)
) +
  geom_point(na.rm = TRUE) +
  facet_grid(species ~ island)
```



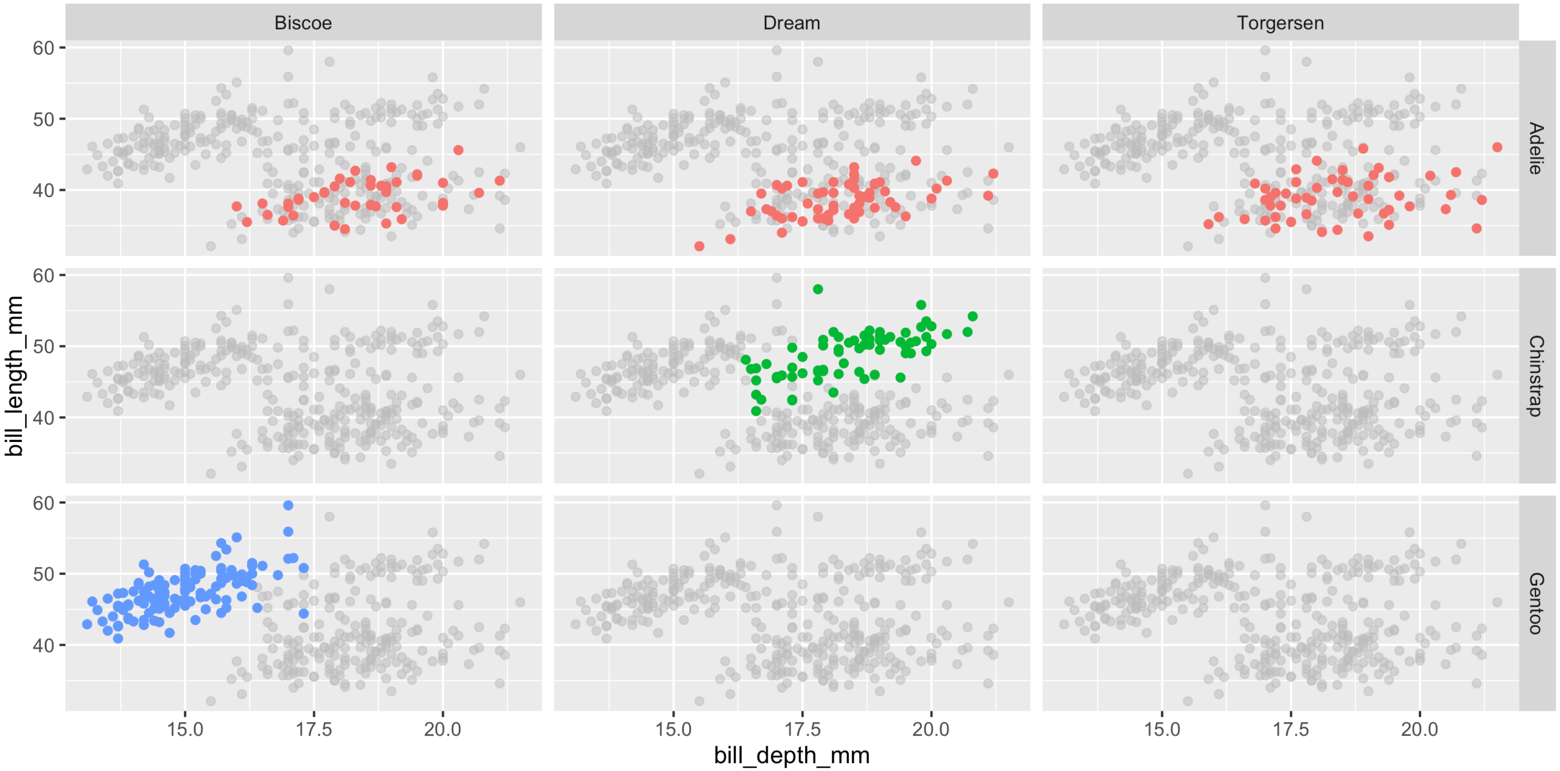
Hiding legend elements

```r
ggplot(
  penguins, aes(x = bill_depth_mm, y = bill_length_mm, color = species)
) +
  geom_point(na.rm = TRUE) +
  facet_grid(species ~ island) +
  guides(color = "none")
```



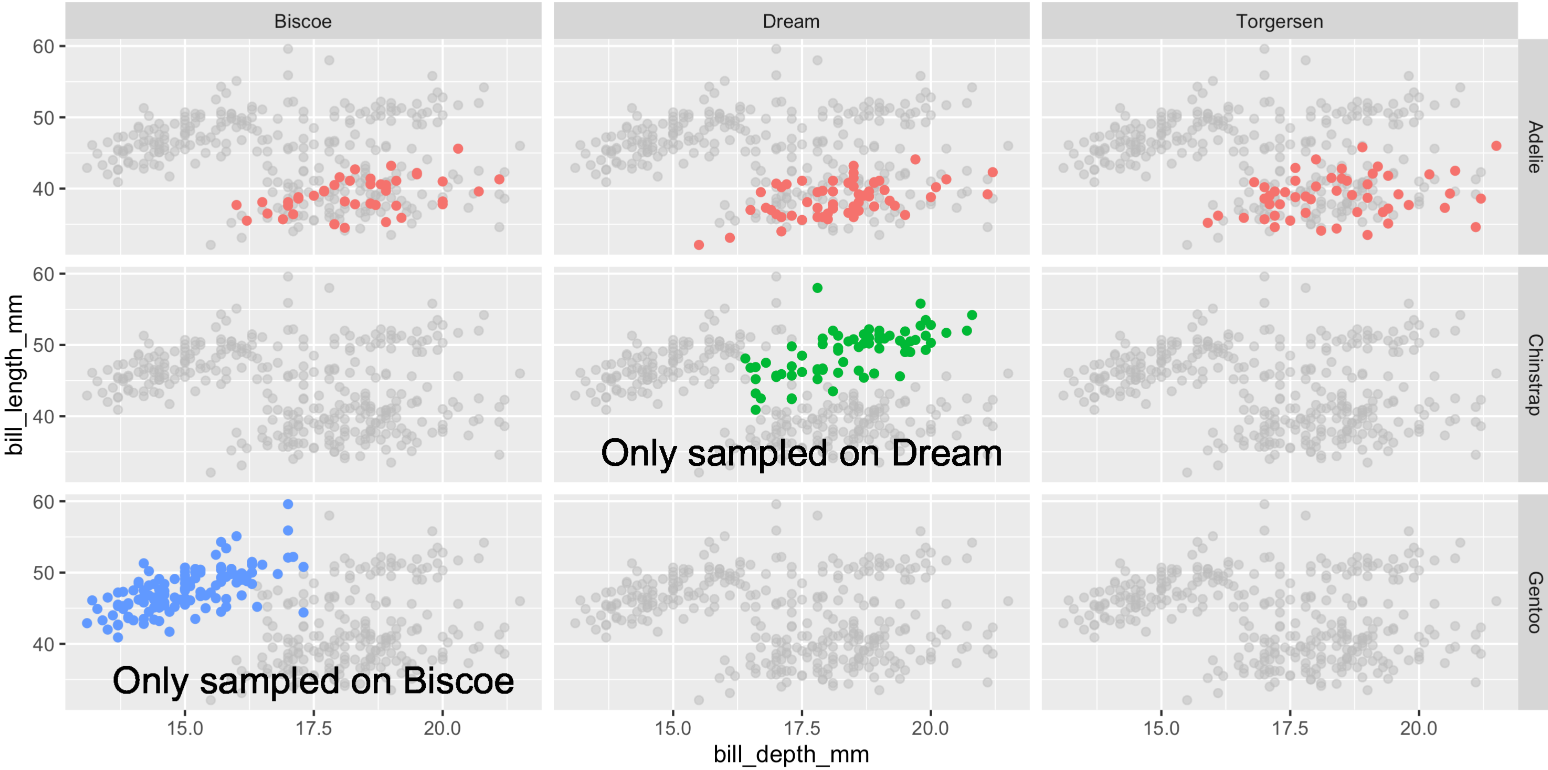
Facet layout - context

```
ggplot(
  penguins, aes(x = bill_depth_mm, y = bill_length_mm, color = species)
) +
  geom_point(color = "grey", alpha = 0.5, na.rm = TRUE, layout = "fixed") +
  geom_point(na.rm = TRUE) +
  facet_grid(species ~ island) +
  guides(color = "none")
```



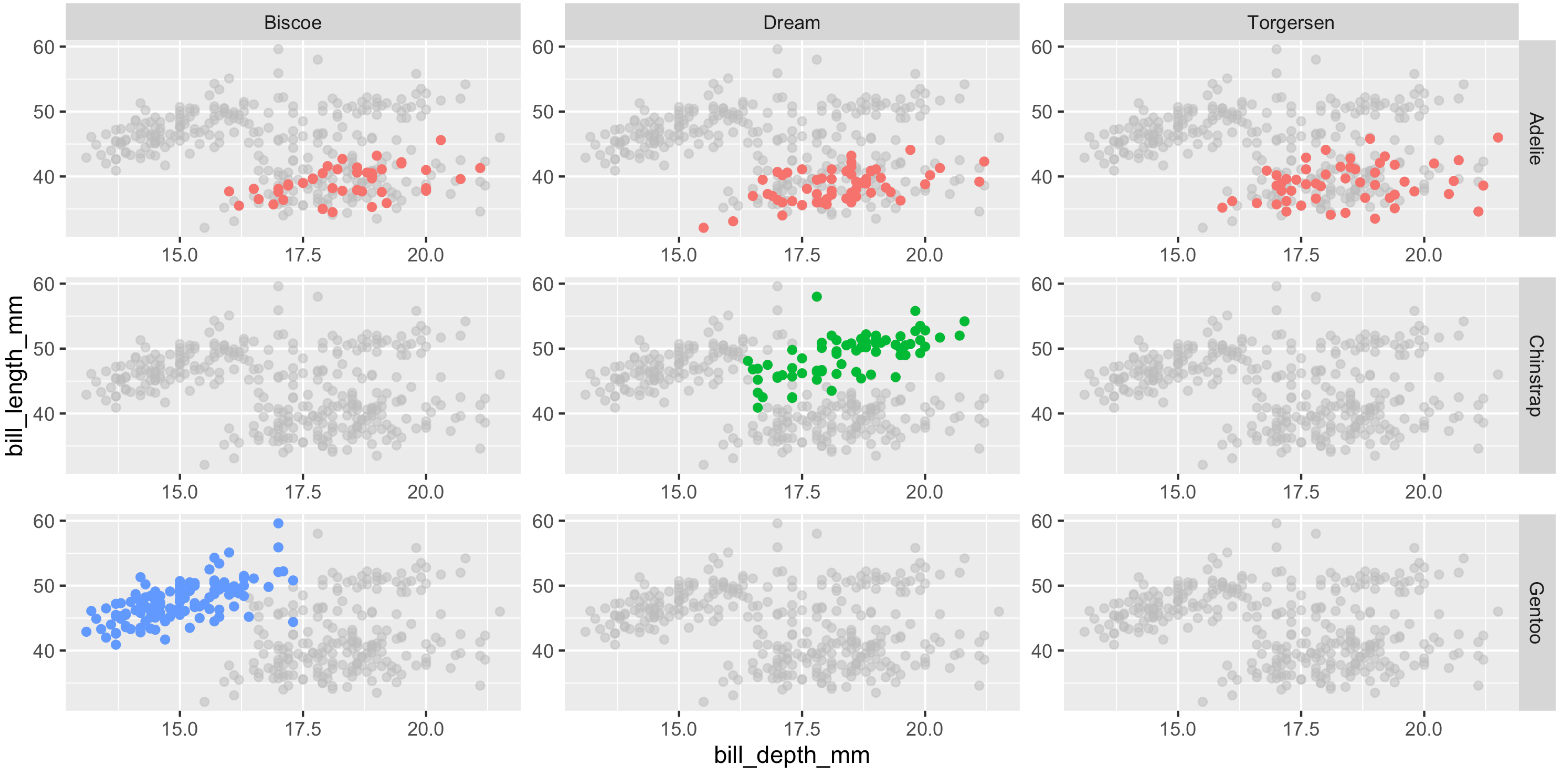
Facet layout - annotation

```r
ggplot(
  penguins, aes(x = bill_depth_mm, y = bill_length_mm, color = species)
) +
  geom_point(color = "grey", alpha = 0.5, na.rm = TRUE, layout = "fixed") +
  geom_point(na.rm = TRUE) +
  facet_grid(species ~ island) +
  guides(color = "none") +
  geom_text(
    x = 17.5, y = 35, label = "Only sampled on Dream", size = 6, color = "black",
    layout = 5
  ) +
  geom_text(
    x = 17.5, y = 35, label = "Only sampled on Biscoe", size = 6, color = "black",
    layout = 7
  )
```


Biscoe
Dream
Torgersen
Adelie
Chinstrap
Gentoo
bill_length_mm
bill_depth_mm
60
50
40
15.0
17.5
20.0
Only sampled on Dream
Only sampled on Biscoe

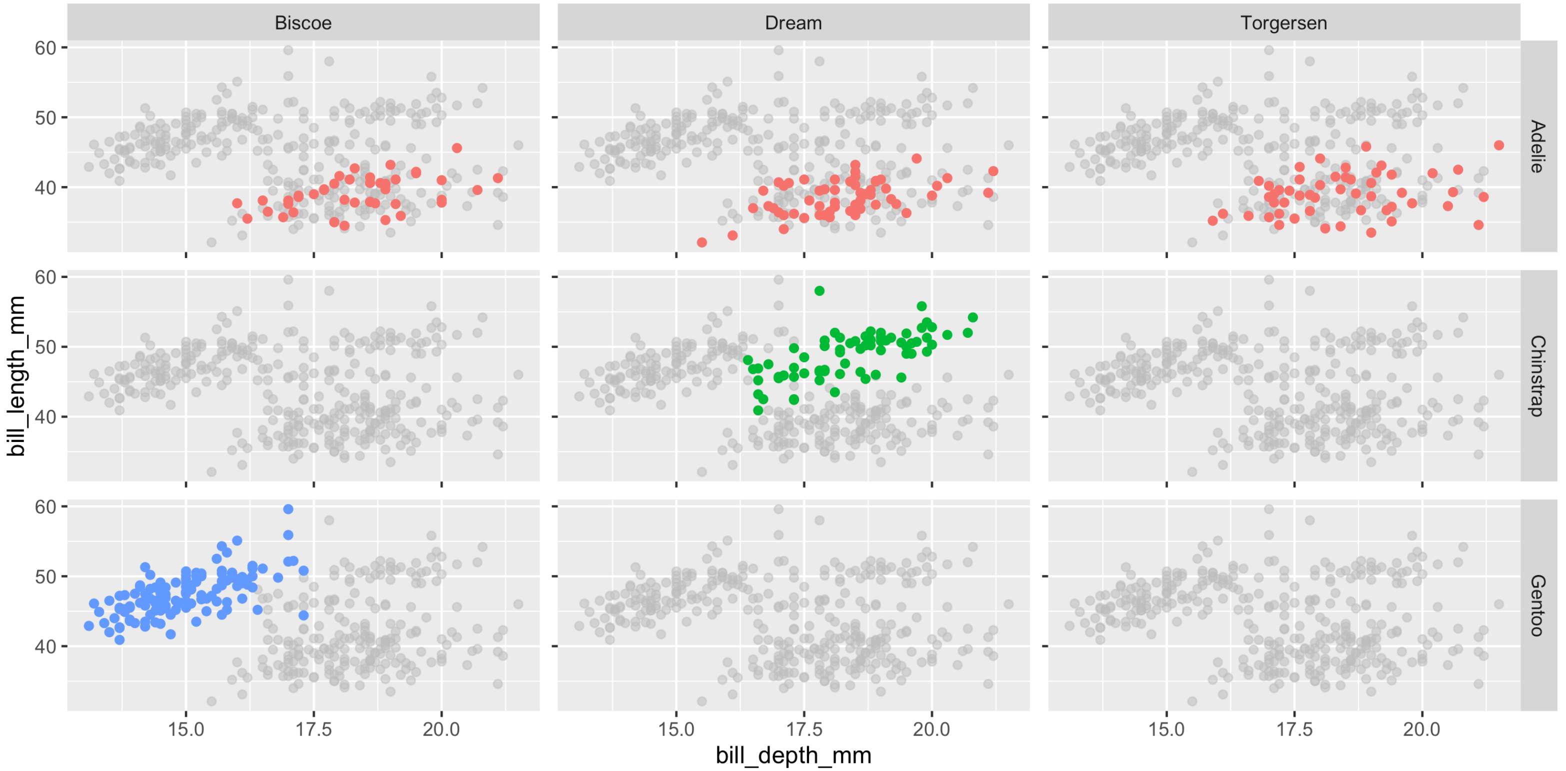
Facet axes

```
ggplot(
  penguins, aes(x = bill_depth_mm, y = bill_length_mm, color = species)
) +
  geom_point(color = "grey", alpha = 0.5, na.rm = TRUE, layout = "fixed") +
  geom_point(na.rm = TRUE) +
  facet_grid(species ~ island, axes = "all") +
  guides(color = "none")
```



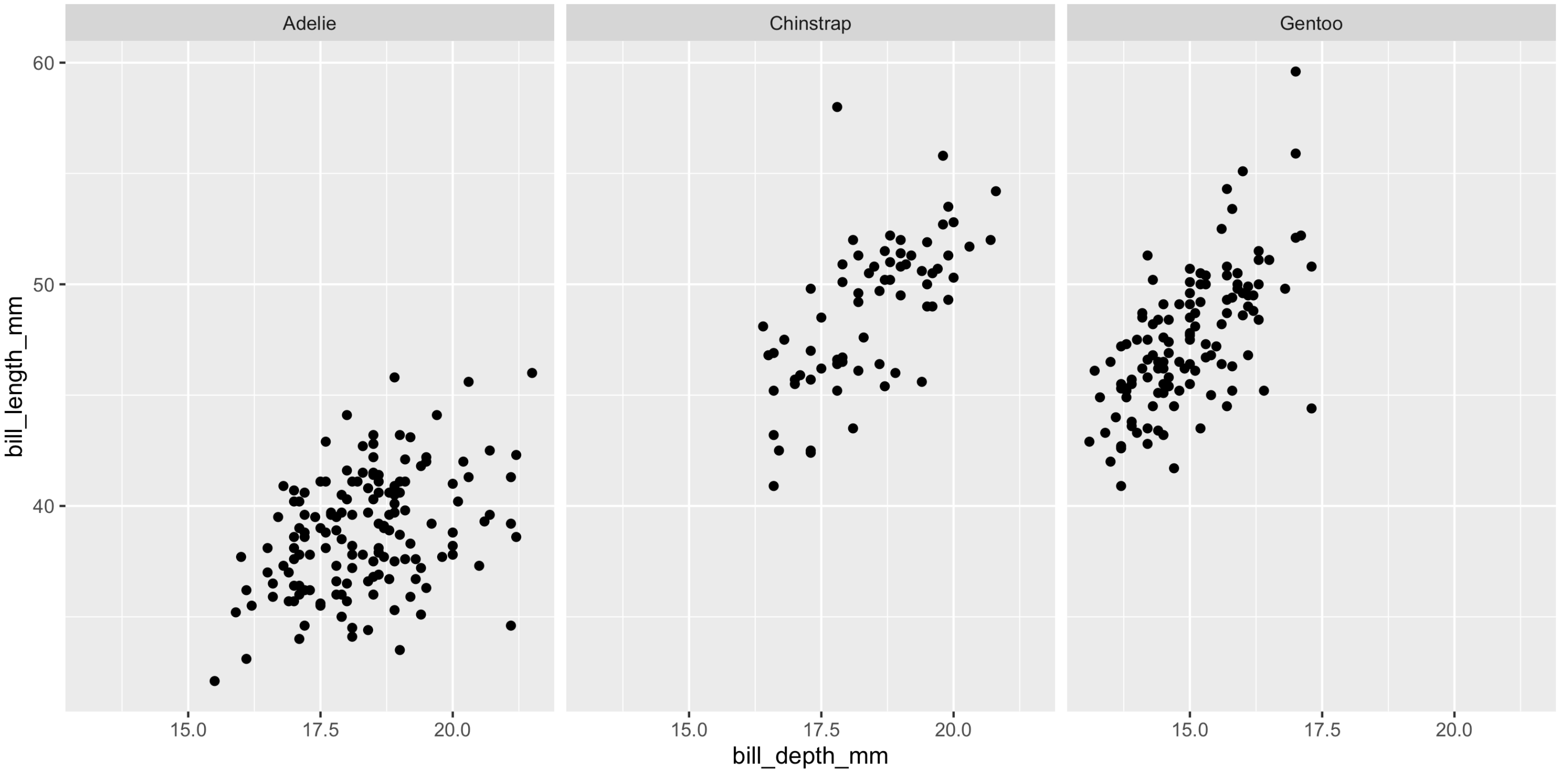
Facet axes - labels

```
ggplot(
  penguins, aes(x = bill_depth_mm, y = bill_length_mm, color = species)
) +
  geom_point(color = "grey", alpha = 0.5, na.rm = TRUE, layout = "fixed") +
  geom_point(na.rm = TRUE) +
  facet_grid(species ~ island, axes = "all", axis.labels = "margins") +
  guides(color = "none")
```



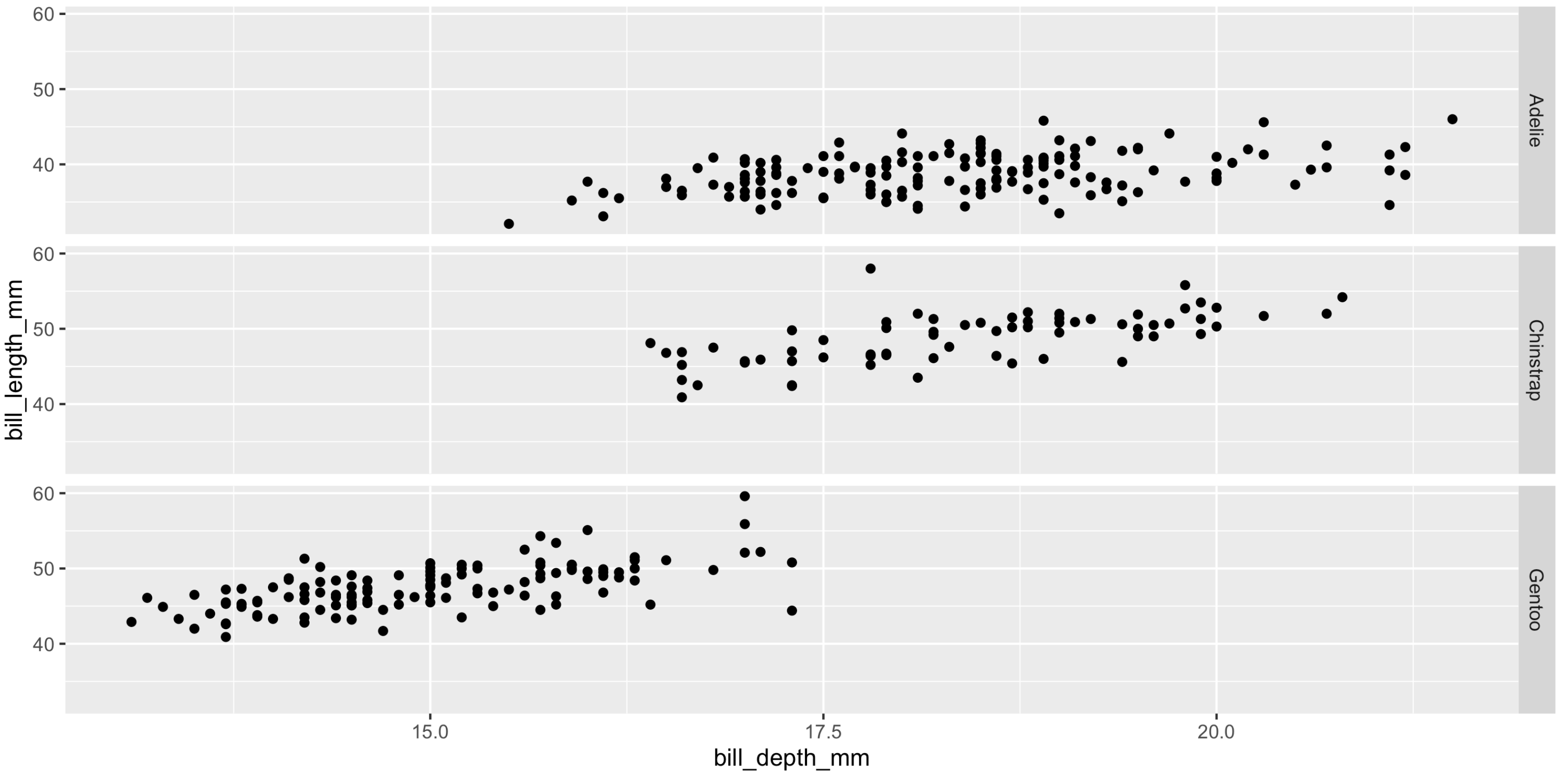
facet_grid (columns)

```
ggplot(
  penguins, aes(x = bill_depth_mm, y = bill_length_mm)
) +
  geom_point(na.rm = TRUE) +
  facet_grid(~ species)
```



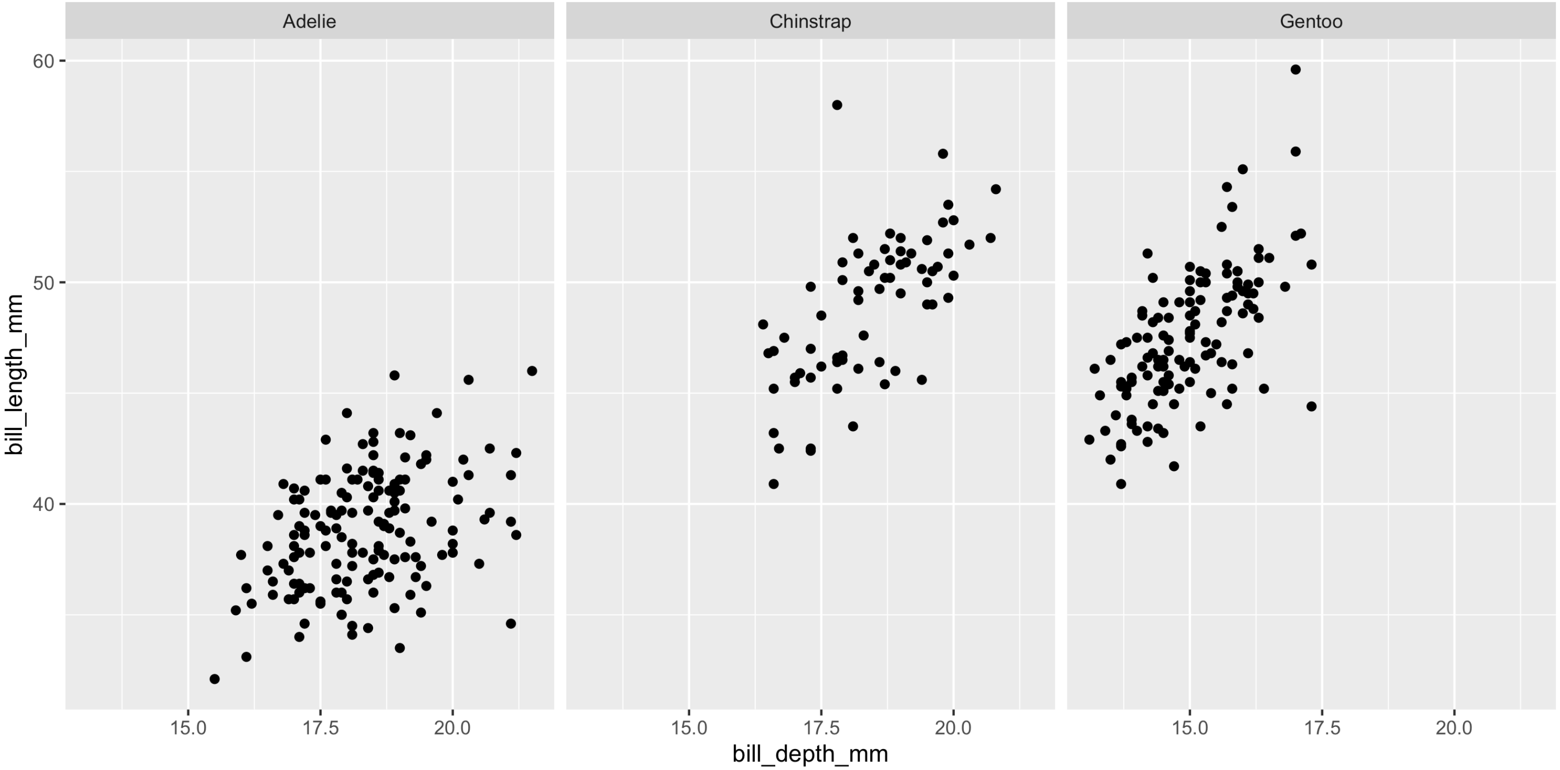
facet_grid (rows)

```r
ggplot(
  penguins, aes(x = bill_depth_mm, y = bill_length_mm)
) +
  geom_point(na.rm = TRUE) +
  facet_grid(species ~ .)
```



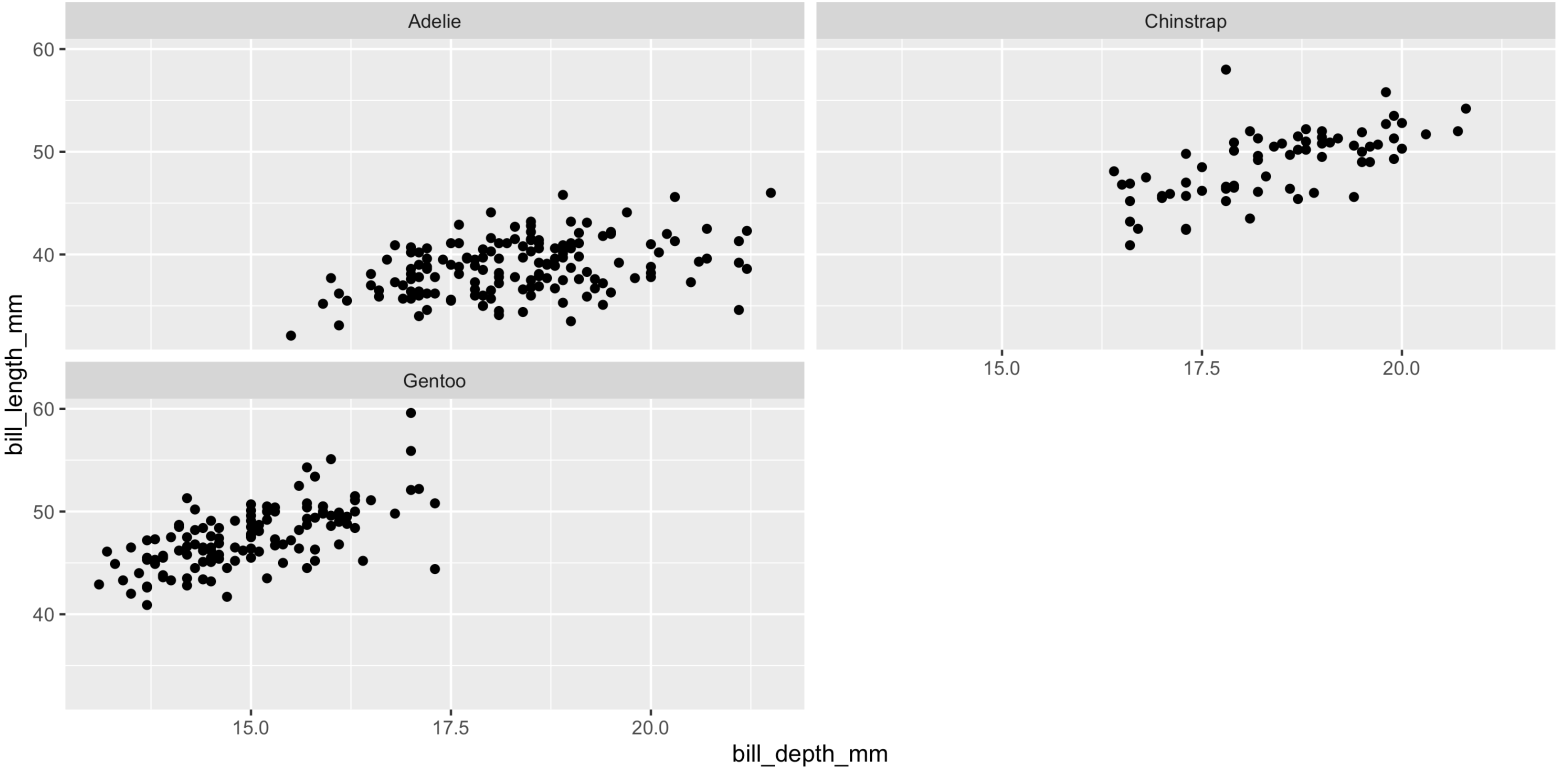
facet_wrap

```r
ggplot(
  penguins, aes(x = bill_depth_mm, y = bill_length_mm)
) +
  geom_point(na.rm = TRUE) +
  facet_wrap(~ species)
```



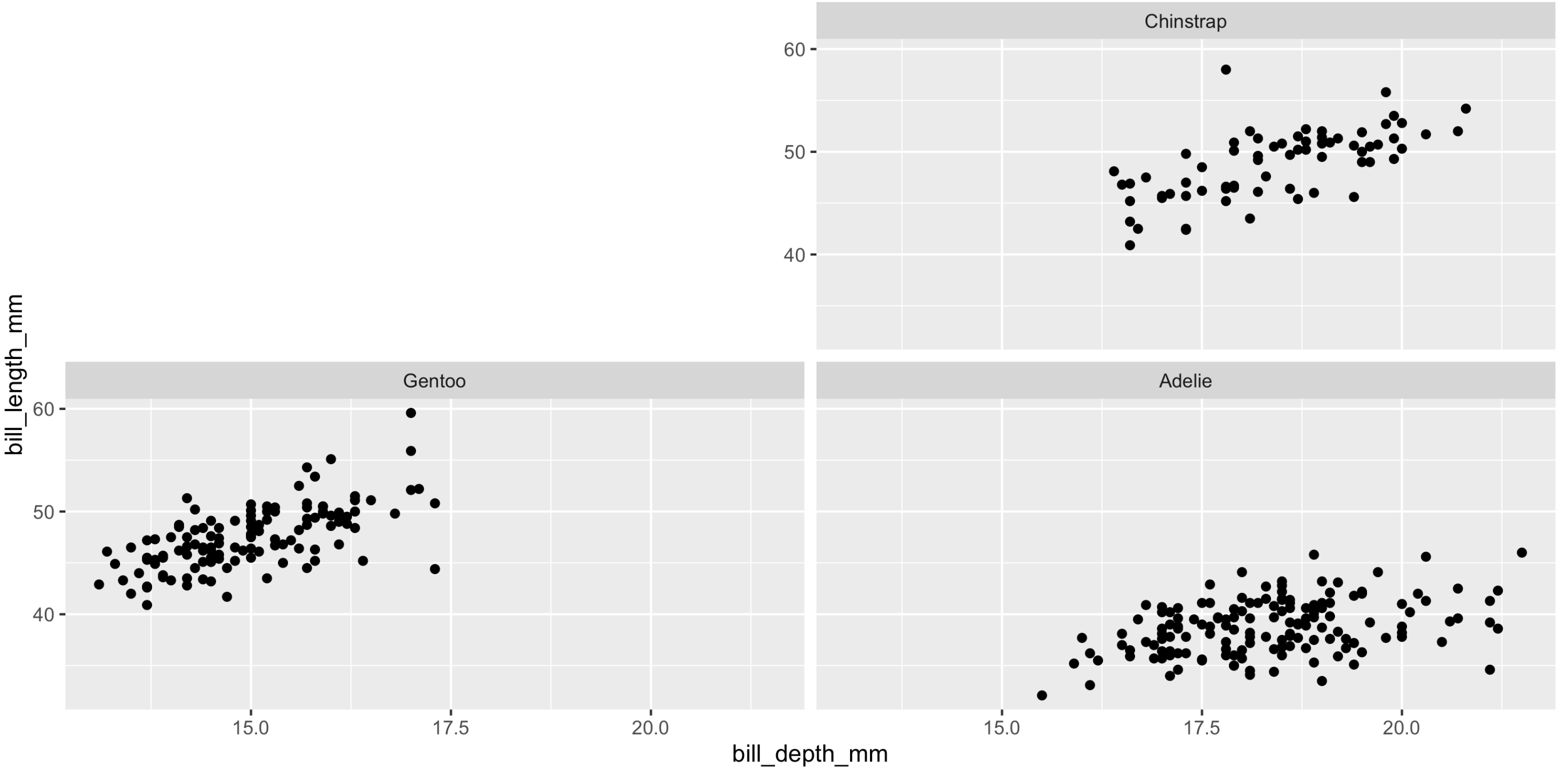
facet_wrap

```r
ggplot(
  penguins, aes(x = bill_depth_mm, y = bill_length_mm)
) +
  geom_point(na.rm = TRUE) +
  facet_wrap(~ species, ncol = 2)
```



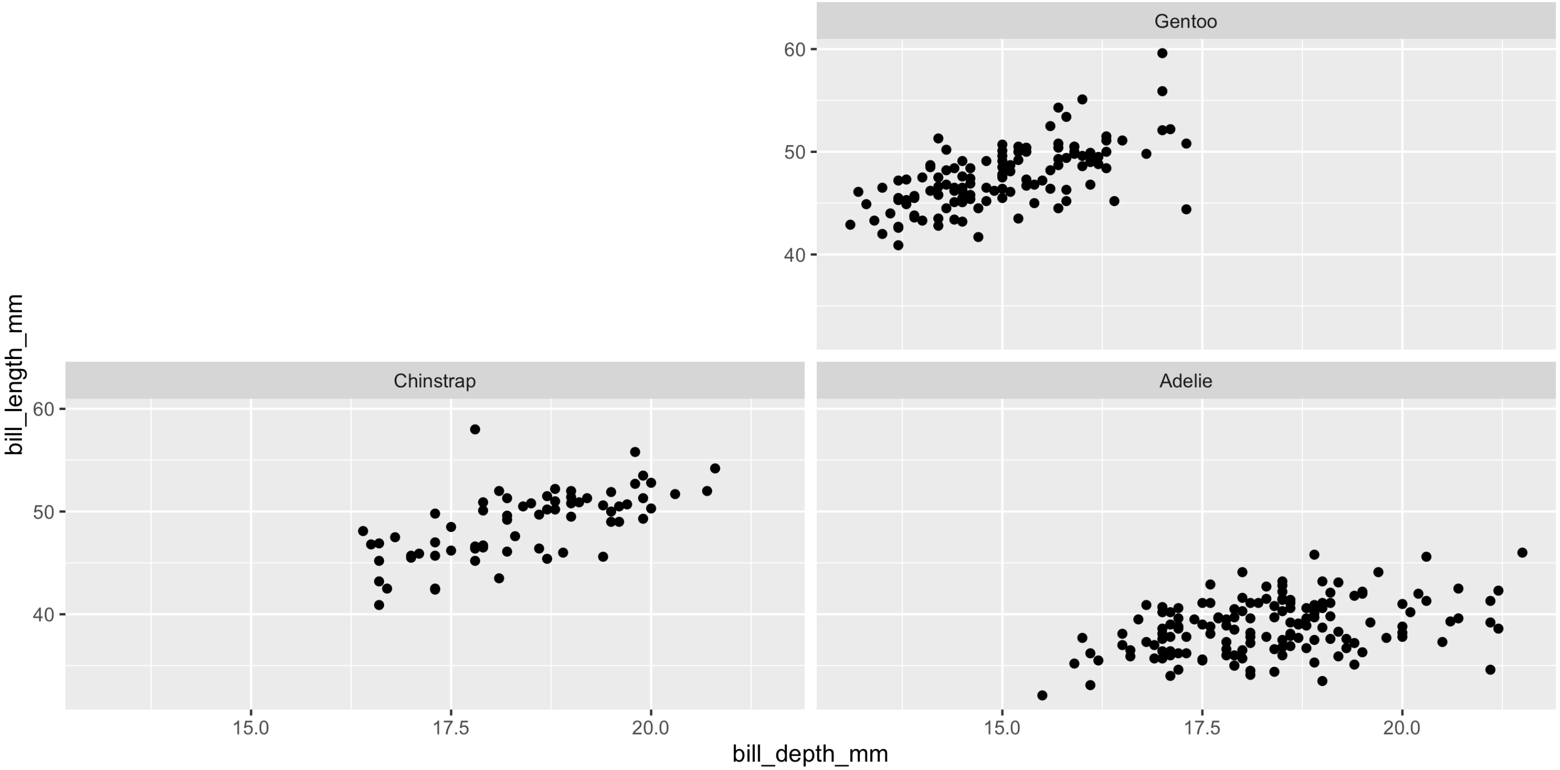
facet_wrap - direction

```
ggplot(
  penguins, aes(x = bill_depth_mm, y = bill_length_mm)
) +
  geom_point(na.rm = TRUE) +
  facet_wrap(~ species, ncol = 2, dir = "br")
```



facet_wrap - direction

```r
ggplot(
  penguins, aes(x = bill_depth_mm, y = bill_length_mm)
) +
  geom_point(na.rm = TRUE) +
  facet_wrap(~ species, ncol = 2, dir = "rb")
```



Learning more

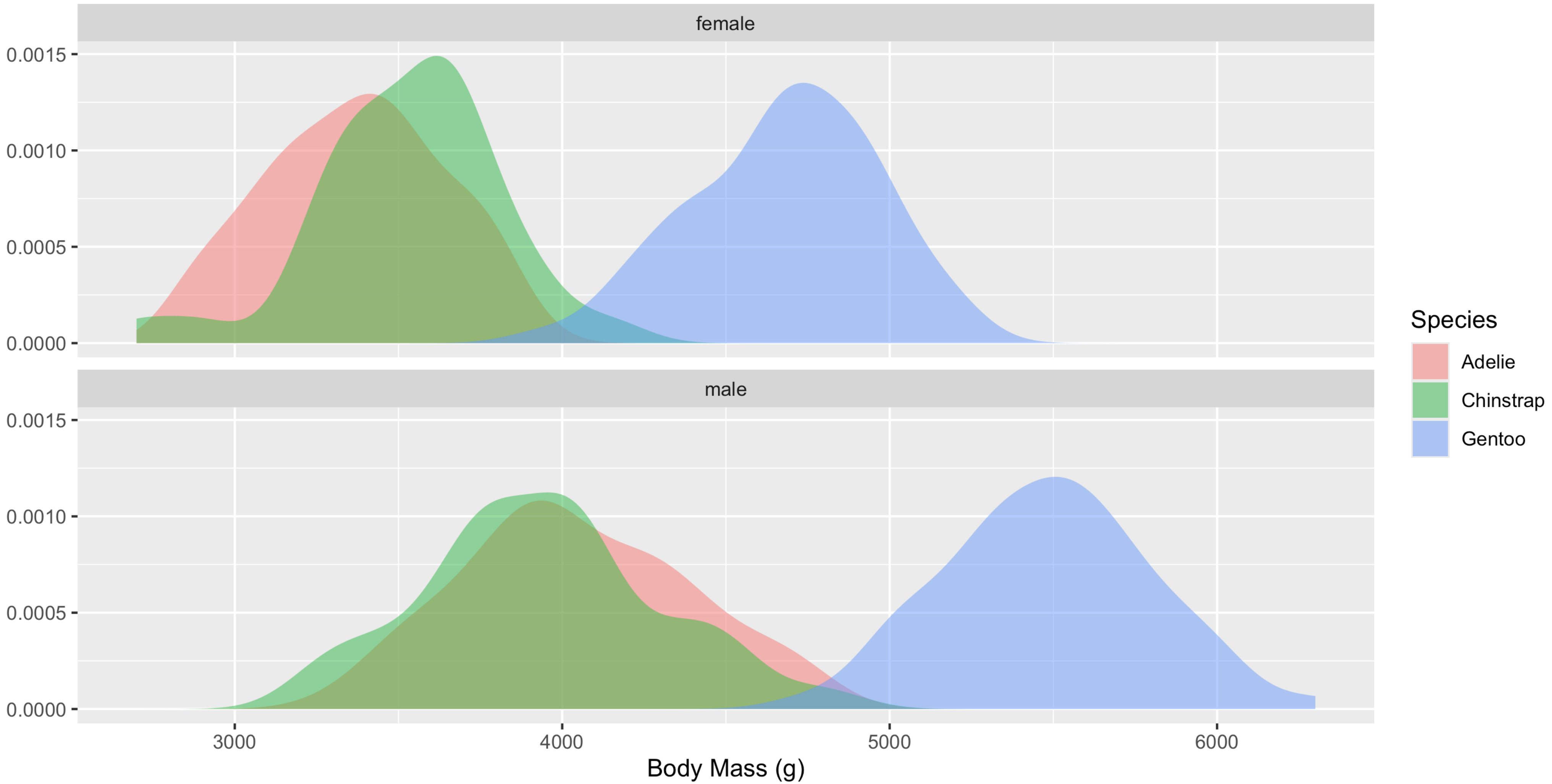
geom tour

ggplot2.tidyverse.org/reference/index.html

Exercises

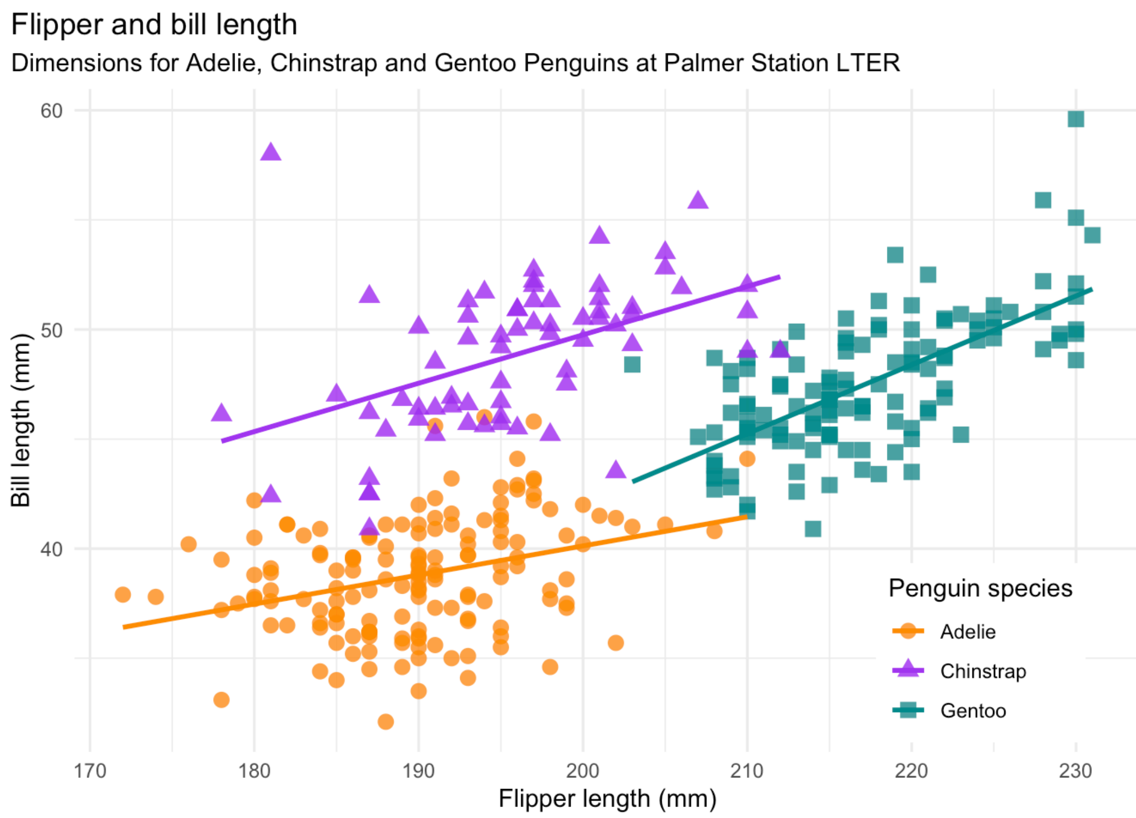
Exercise 1

Recreate, as faithfully as possible, the following plot using ggplot2 and the `penguins` data.



Exercise 2

Recreate, as faithfully as possible, the following plot from the `palmerpenguin` package README in ggplot2.



See the `palmerpenguins` pkgdown site